# **STAT 27850:** Project 1

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# 1 Introduction

The core objective of this study is to move from literal data interests to a methodological inquiry regarding the stability and validity of inference in high-dimensional observational data. We pose the following central question: **When testing for neighborhood-level changes in taxi usage between 2014 and 2019, how do different strategies for hypothesis family design, multiplicity reduction, and null calibration trade off the control of false discoveries against the detection of stable, reproducible signals?** To provide a comprehensive answer, we decompose this core question into three specific methodological pillars:

1. Hypothesis Family Design and Multiplicity: How does the granularity of the "family" (e.g., testing neighborhoods marginally versus testing *neighborhood* $\times$ *time* interactions) shift the discovery set? We compare a "Neighborhood-only" family ($m \approx 231$ tests) against a "Context-dependent" family ($m \approx 693$ tests) to evaluate the cost of increasing scientific resolution.
2. Inferential Levers via Data Reduction: To what extent does sample-size thresholding ($n_{min}$) act as a stabilizer for $p$-values? We quantify the trade-off between "power" (retaining more neighborhoods) and "reliability" (filtering low-volume cells that produce erratic tail-driven noise).
3. Calibration under Dependence: In the presence of spatial and temporal correlation, do standard asymptotic/parametric tests (e.g., difference in proportions for rush-hour timing) inflate discovery rates compared to permutation-based null distributions? We aim to determine if permutation engines provide a more "honest" assessment of significance when the independence assumption is compromised.

Finally, we validate these pipelines through Placebo and Stability checks. By applying our methodology to "fake" rush-hour windows and sub-sampled data, we identify which discoveries are robust behavioral shifts and which are mere artifacts of the statistical pipeline chosen.

# 2 Data Processing & Study Design

## 2.1 Data Cleaning

The raw dataset initially comprises 400,000 observations, representing a random sample of complete records from 2014 and 2019. To ensure the inferential validity of our testbed, we implemented a hierarchical cleaning pipeline designed to eliminate corrupted data, physical impossibilities, and extreme outliers that could distort asymptotic variability.

1. Temporal and Distance Validity: We removed 591 records with non-positive trip durations (trip_seconds), non-positive distances (trip_miles), or negative cost components (fares, tips, or extras).
2. Outlier Mitigation: To prevent extreme tail events from skewing our neighborhood-level summaries, we capped records at established physical and economic limits:
   (a) Maximum distance: 100 miles
   (b) Maximum duration: 4 hours
   (c) Maximum total cost/fare: $500

   This step removed 168 extreme outliers, resulting in a final analytic sample of 399,241 rides.
3. Handling Missingness: Exploratory analysis revealed significant missingness in pickup_census_tract (101,441 missing values). Consequently, we utilize pickup_community_area (0% missing) as our primary spatial unit to avoid the selection bias inherent in discarding half the dataset.

## 2.2 Covariate Control & Confounders

The assignment warns that apparent behavioral changes may be confounded by non-behavioral factors such as fare increases, traffic changes, or shifts in taxi company operations. Our outcome and covariate choices are designed to address this concern from multiple perspectives.

**Outcome design to handle confounder.** Each of our three outcomes is defined to be inherently robust to common confounders:

- **Rush-hour share** ($Z_{\text{rush}}$) tests the proportion of rides occurring during rush hours, not the count. This is invariant to changes in overall taxi demand: if a neighborhood loses 50% of its rides to rideshare but the remaining riders maintain the same temporal pattern, the proportion is unchanged. Similarly, fare increases do not affect when rides occur.
- **Cash payment share** ($Z_{\text{cash}}$) reflects payment preference instead of price level. Whether a customer pays cash or card is independent of the fare amount, so this outcome is unconfounded by fare inflation.
- **Tipping propensity** ($Z_{\text{tip}}$) uses a binary indicator (any tip vs. no tip) among non-cash rides only. The binary formulation avoids confounding with fare-driven changes in tip amounts. Restricting to non-cash payments avoids the selection bias from unrecorded cash tips: cash rides mechanically show tip = 0 in the data regardless of whether a cash tip was actually given.

**Covariate adjustment in the logistic model.** For the covariate-adjusted p-value engine in Section 5, we fit per-neighborhood logistic regressions of the form

$$\operatorname{logit} \mathbb{P}(Z = 1) = \beta_0 + \beta_1 \mathbf{1}\{\text{year} = 2019\} + \gamma_1 \,\text{month} + \gamma_2 \mathbf{1}\{\text{weekend}\},$$

where the coefficient of interest is $\beta_1$ (the year effect). The covariates `month` and `is_weekend` absorb seasonal and day-of-week variation that could otherwise be aliased with the year effect. For instance, if the 2014 and 2019 samples over-represent different months, the unadjusted $z$-test could mistake seasonal variation for a year effect, which the logistic model helps control for.

**Residual confounders.** Several potential confounders cannot be addressed with the available data. The rise of rideshare services (Uber, Lyft) between 2014 and 2019 likely altered the composition of taxi riders. Specifically, if price-sensitive or late-night riders switched to rideshare, the remaining taxi riders may differ systematically from the 2014 population.

**Note on Notation** $p_a^{(y)}$ (or $p_{a,y}$) denotes the population proportion of outcome $Z$ in neighborhood $a$ during year $y$, and $\hat{p}_a^{(y)}$ its sample estimate. Where the outcome needs to be distinguished, we write $p_a^{(y)}(\text{rush})$, $p_a^{(y)}(\text{cash})$, or $p_a^{(y)}(\text{tip})$. If modifications to superscript and subscript usage, they are explained in corresponding sections.

# 3 Exploratory Data Analysis

Before conducting formal hypothesis tests, we examine the structure of the cleaned dataset to characterize baseline heterogeneity, assess the scale of observable year-over-year shifts, and identify where sampling variability may obscure or exaggerate apparent behavioral change. The goal of this section, rather than drawing inferential conclusions, is to understand the geometry (effect size, variance heterogeneity, reality constraints) and motivate our testing framework, developing intuition and calibration for Section 4.

## 3.1 Hypothesis Family Design

Our analysis is organized around three binary outcomes that capture distinct dimensions of customer behavior while avoiding confounding with fare inflation, traffic changes, or other non-behavioral factors:

1. **Rush-hour usage share** ($Z_{\text{rush}}$): whether a trip occurs during rush hours (7:00–9:59 AM or 4:00–6:59 PM Chicago local time). Testing the *proportion* rather than the count ensures robustness to changes in overall taxi demand.
2. **Cash payment share** ($Z_{\text{cash}}$): whether the customer pays with cash. This reflects payment preference and is not confounded by price-level changes.
3. **Tipping propensity** ($Z_{\text{tip}}$): among non-cash rides only, whether any tip is left. Restricting to non-cash payments avoids the selection bias from unrecorded cash tips; using the binary indicator rather than tip amount avoids confounding with fare increases.

For each outcome, we define two hypothesis families at different levels of granularity.

- **Family 1 (Neighborhood-only)** tests one hypothesis per pickup community area: whether the population proportion of the outcome changed between 2014 and 2019.
- **Family 2 (Neighborhood $\times$ Time-bin)** crosses each neighborhood with five time-of-day bins (late night 00:00–05:59, morning rush 06:00–09:59, midday 10:00–15:59, evening rush 16:00–18:59, night 19:00–23:59), testing whether the outcome changed within each cell.

Family 2 increases scientific resolution at the cost of a larger multiplicity burden and smaller per-cell sample sizes. The trade-off between these two family designs is itself a central object of study in this project. We require a minimum of $n_{\min}$ rides per cell per year for a hypothesis to be included, treating $n_{\min}$ as a tunable methodological lever whose effect on discoveries we quantify in Section 4.4.

## 3.2 Temporal Patterns (2014 vs 2019)

### 3.2.1 Hourly ride distribution

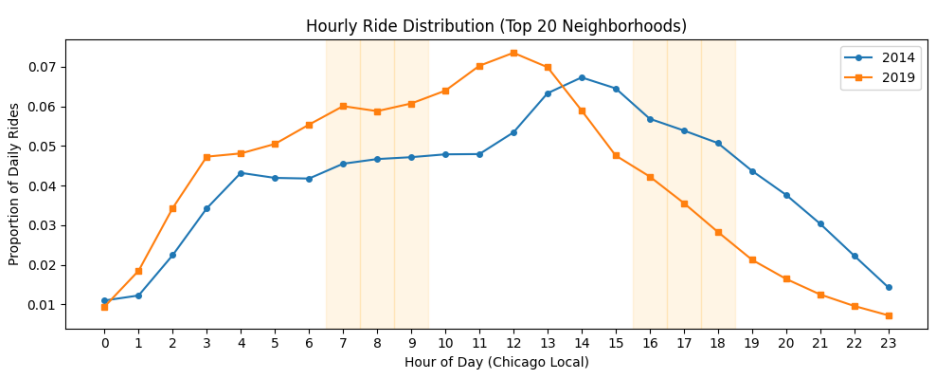



Figure 1: Hourly ride distribution by year. Rush-hour windows (7–9 AM, 4–6 PM) are shaded.

The above figure overlays the hourly proportion of rides (normalized within each year) for 2014 and 2019 across the top 20 neighborhoods, with rush-hour windows shaded. The two distributions differ markedly. In 2014, ride volume was relatively flat from mid-morning through early evening, with a broad plateau between hours 10 and 18. By 2019, the distribution had shifted substantially toward earlier hours: the peak moved from the early afternoon (hours 12–14) to mid-morning (hours 10–12), and the evening decline steepened. Rides after 7:00 PM dropped sharply in 2019 relative to 2014, while early-morning and morning-rush volumes increased.

This aggregate temporal shift has two important implications for our analysis. First, it confirms that meaningful behavioral changes occurred between the two years, motivating formal testing. Second, it explains why our initial placebo tests (which tested arbitrary hour windows) failed: the entire hourly distribution shifted, so any fixed set of hours can show a significant year-over-year change. A valid placebo must therefore break the year structure itself, not merely relocate the hour window (see Section 5).

### 3.2.2 Time-bin composition shifts

To examine temporal changes at a finer spatial resolution, Figure 3.2 displays the change in time-bin share (2019 minus 2014) for each neighborhood. Several patterns emerge consistently across neighborhoods: late-night and morning-rush shares increased in 2019 (warm colors in the left two columns), while evening-rush and night shares decreased (cool colors in the right two columns). The midday bin shows mixed but generally small changes.

The consistency of the pattern across neighborhoods suggests a systematic temporal redistribution of taxi demand rather than idiosyncratic neighborhood-level noise. Neighborhoods 41, 5, and 7 show particularly large increases in late-night and morning-rush shares, while neighborhoods 4, 5, and 7 exhibit the largest declines in the night bin. This heterogeneity across neighborhoods — some showing shifts an order of magnitude larger than others — motivates testing at the neighborhood level rather than relying solely on an aggregate test.

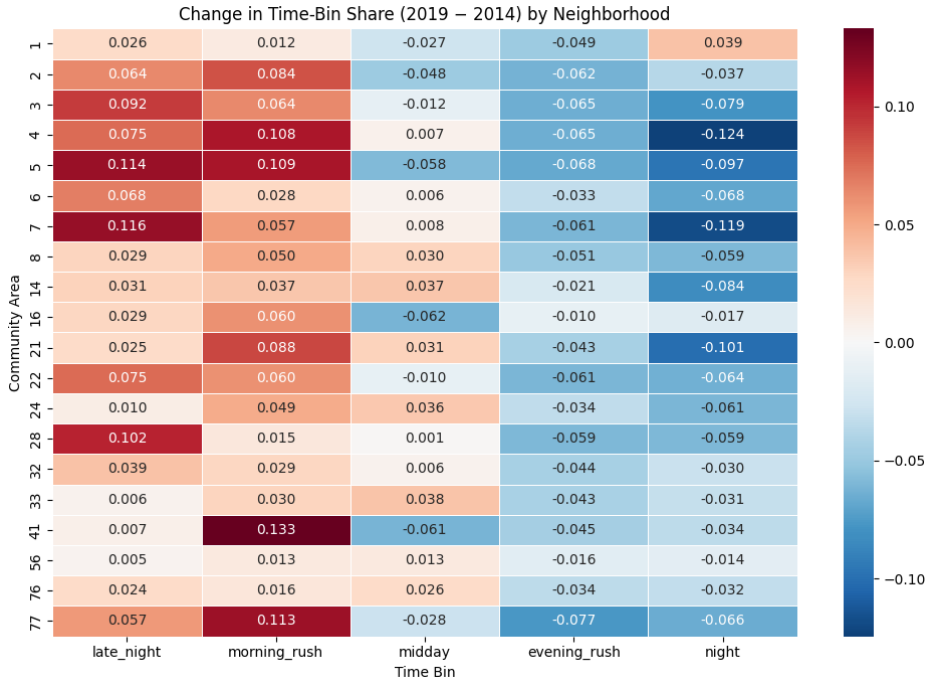



Figure 2: Change in time-bin share (2019  2014) by neighborhood. Rows are community areas; columns are the five time-of-day bins.

## 3.3 Spatial Patterns

### 3.3.1 Ride volume

Figure displays ride counts by neighborhood and year (left panel) and percent change (right panel). The landscape of taxi demand changed dramatically between 2014 and 2019. Community area 8 (Near North Side / Loop) remained the dominant pickup location with stable volume (∼64,000 rides in both years). However, several neighborhoods experienced large and divergent shifts: area 76 (O'Hare) grew by 85%, area 32 (Loop) grew by 45%, while areas 24 (West Town), 22 (Logan Square), and 5 (North Center) each declined by 75% or more.

These volume changes are significant for our methodology in two ways. First, neighborhoods with sharply declining volume will have smaller 2019 cell sizes, reducing power and potentially failing the $n_{\min}$ threshold — making the choice of $n_{\min}$ consequential. Second, volume changes themselves are *not* the behavioral outcome we test (our hypotheses concern proportions, not counts), but they indicate that the competitive landscape for taxis shifted unevenly across Chicago, providing context for interpreting any detected behavioral changes.

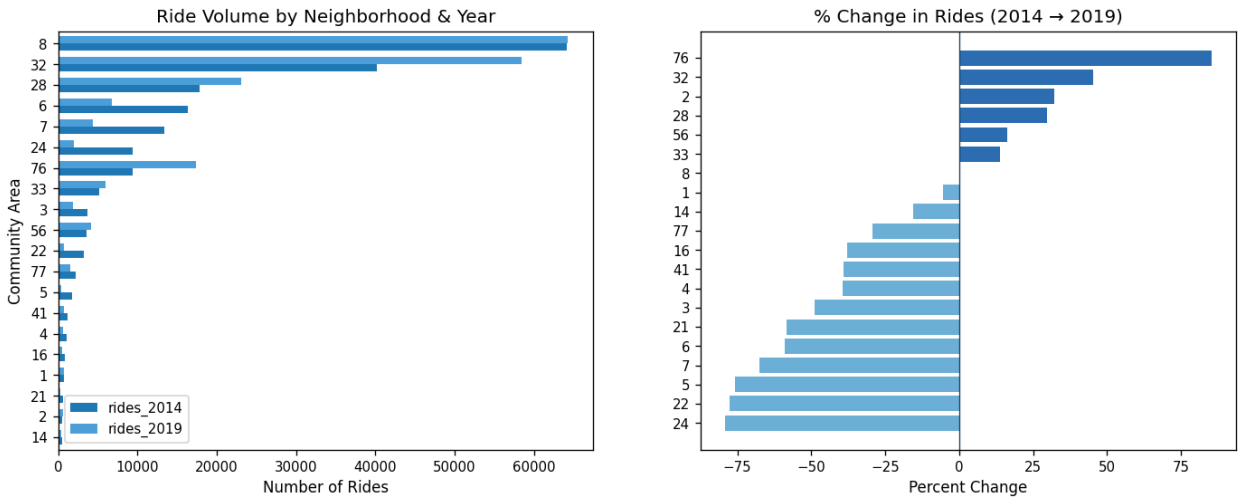



Figure 3: Ride volume by neighborhood and year (left) and percent change (right).

### 3.3.2 Rush-hour share

Figure shows the change in rush-hour proportion (2019 − 2014) for each neighborhood. Most neighborhoods experienced a modest decline in rush-hour share (1–3 percentage points), consistent with the aggregate temporal shift toward earlier hours noted above. However, several neighborhoods buck this trend: areas 41 (Hyde Park) and 16 (Irving Park) saw their rush-hour share *increase* by 5.6 and 8.5 percentage points respectively, suggesting neighborhood-specific behavioral shifts beyond the citywide temporal redistribution.

The effect sizes are modest — the largest absolute change is 8.5 percentage points — which places a premium on the statistical power of our tests. The heterogeneity in both sign and magnitude across neighborhoods confirms that a single aggregate test would be insufficient; neighborhood-level testing is essential to characterize the spatial structure of these shifts.

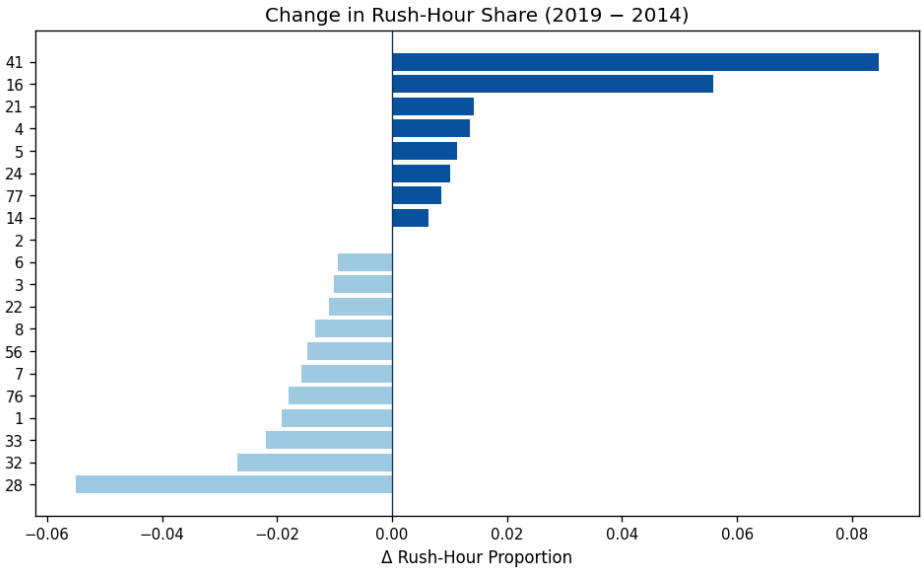



Figure 4: Change in rush-hour share (2019 - 2014) by neighborhood.

### 3.4 Key EDA Takeaways

Before turning to formal inference, the exploratory patterns across time, space, and payment behavior reveal several structural features of the data that are directly relevant for our multiple testing framework.

To begin with, behavioral change between 2014 and 2019 is not uniform across dimensions. Particularly, the hourly distribution shows a clear temporal redistribution of rides, with demand shifting toward earlier hours and away from late evening. At the neighborhood level, time-bin composition changes exhibit systematic structure: increases in the morning bins are typically offset by declines in the evening and night bins. Since time-bin shares sum to one within each neighborhood-year, these shifts are inherently compositional rather than independent. This compositional reality constraint implies statistical dependence across hypotheses within a neighborhood when testing bin-specific changes (Family 2), a feature that standard multiple testing procedures may not fully account for.

Secondly, sample size heterogeneity is pronounced, in the sense that ride volume differs dramatically across neighborhoods and volume changes between 2014 and 2019 are uneven. Although are hypotheses concern proportions rather than counts / absolute magnitude, sample size directly determines the precision of the binomial estimators underlying each test. Two neighborhoods exhibiting identical percentage changes may yield very different p-values purely due to differences in cell size. This variance heterogeneity suggests that the distribution of discoveries will depend not only on effect size but also on our inclusion criteria. In particular, the minimum cell-size threshold $n_{min}$ acts as a methodological lever: raising $n_{min}$ reduces the variance and the total number of hypotheses to be tested, potentially increasing stability at the cost of excluding smaller neighborhoods.

In addition, resolution and multiplicity trade off sharply. Family 1 neighborhood level tests involve a moderate number of hypotheses and relatively large sample sizes per unit; Family 2 neighborhood $\times$ time-bin increases scientific resolution by a factor of five but simultaneously increases multiplicity and reduces per-cell counts. The exploratory heatmap makes clear that many time-bin shifts are small in magnitude; whether these represent genuine behavioral change or sampling fluctuation cannot be determined visually.

Last but not least, the EDA highlights risk of over-interpreting extreme percent changes in low-volume neighborhoods. Several neighborhoods exhibit very large relative growth or decline in ride counts, driven by small 2014 baselines.Such instability reinforces our decision to test binary outcomes (proportions) rather than raw counts, but it also cautions that even proportion changes may be noisy when denominators are small. This further motivates systematic sensitivity analysis with respect to inclusion thresholds and correction procedures.

# 4 Hypothesis Testing Framework & Results

We use Chicago's pickup community areas as the spatial unit. The city contains 77 community areas, but ride volume is highly concentrated: the top 20 neighborhoods account for over 97% of rides in our dataset. This suggests the presence of a $n_{\min}$, where we only look at neighborhoods that have number of rides greater than $n_{\min}$, to be helpful. This restriction serves a dual methodological purpose:

1. **Statistical stability.** Low-volume neighborhoods produce volatile test statistics and erratic p-values. Restricting to high-volume neighborhoods ensures that each hypothesis is backed by adequate data.
2. **Multiplicity control.** Fewer hypotheses means a less severe correction, preserving power for the neighborhoods where detection is feasible.

For Family 2, we partition the 24-hour day into five time bins based on standard commuter patterns and taxi demand structure:

| Time bin | Hours (Chicago local) | Rationale |
|---|---|---|
| Late night | 00:00–05:59 | Low-demand period |
| Morning rush | 06:00–09:59 | AM commute window |
| Midday | 10:00–15:59 | Off-peak, stable demand |
| Evening rush | 16:00–18:59 | PM commute window |
| Night | 19:00–23:59 | Evening activities |

These bins are chosen to be coarse enough that most cells have adequate sample sizes, yet fine enough to distinguish commute-driven from leisure-driven taxi usage. The rush-hour outcome ($Z_{\text{rush}}$) is defined using hours 7–9 and 16–18 (a subset of the morning rush and evening rush bins), so the Family 2 tests can detect whether rush-hour shifts are concentrated in particular time-of-day contexts. Combining the spatial and temporal units yields two families:

- **Family 1 (Neighborhood-only):** One hypothesis per community area. For outcome $Z$ and neighborhood $a$:
$$H_{0,a} : p_a^{(2014)} = p_a^{(2019)} \qquad \text{vs.} \qquad H_{1,a} : p_a^{(2014)} \neq p_a^{(2019)}.$$
- **Family 2 (Neighborhood $\times$ Time-bin):** One hypothesis per $(a, t)$ cell. For neighborhood $a$ and time bin $t$:
$$H_{0,a,t} : p_{a,t}^{(2014)} = p_{a,t}^{(2019)} \qquad \text{vs.} \qquad H_{1,a,t} : p_{a,t}^{(2014)} \neq p_{a,t}^{(2019)}.$$

In both families, hypotheses are included only if the cell contains at least $n_{\min}$ rides per year, where $n_{\min} = 100$ is our default threshold.

## 4.1 Family 1: Neighborhood-Level Tests

We test the null hypothesis:

$$H_0 : \{\text{There is no change in the proportion of rush-hour in neighborhood } a \text{ across all 77 neighborhoods.}\}$$

In addition, we implement a lower-bound $n_{\min}$, the minimum amount of rides starting from each neighborhood, to remove neighborhoods that don't have too many taxi rides in either 2014 or 2019, and therefore could be affected more by outliers. However, for large $n_{\min}$, this can sometimes induce selection bias, where we are only selecting into large neighborhoods (ie neighborhoods close to airports).

### 4.1.1 Assumptions

1. **(Approximate) Independence Between Neighborhoods**: While this cannot be reasonably assumed, since these neighborhoods are geographically related with one another, this makes our hypothesis test anti-conservative.
2. **Neighborhoods Are Comparable Across Time**: We assume that neighborhoods have not undergone substantial change from 2014-2019 that could otherwise affect taxi data.

### 4.1.2 BH Correction

We first aggregate our data into the following dataframe:

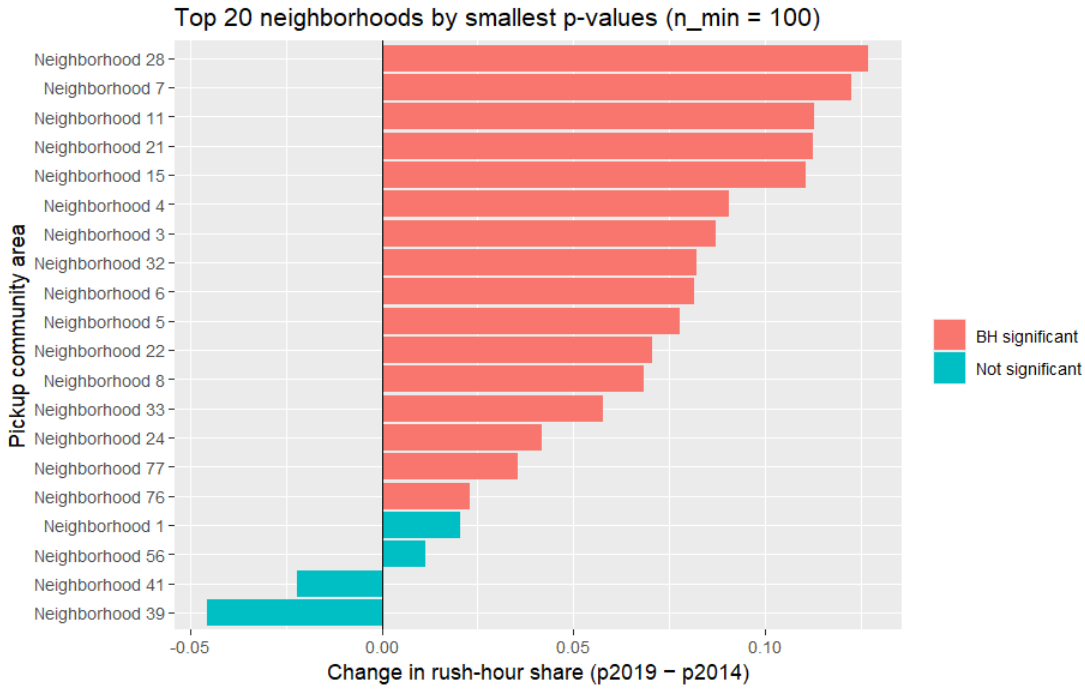
| | pickup_community_area | num_rides_2014 | num_rides_2019 | num_rush_rides_2014 | num_rush_rides_2019 | p_2014 | p_2019 | delta |
|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 760 | 729 | 172 | 180 | 0.22631579 | 0.24691358 | 0.020597791 |
| 2 | 2 | 501 | 663 | 154 | 211 | 0.30738523 | 0.31825038 | 0.010865148 |
| 3 | 3 | 3703 | 1905 | 919 | 639 | 0.24817715 | 0.33543307 | 0.087255917 |
| 4 | 4 | 1071 | 651 | 240 | 205 | 0.22408964 | 0.31490015 | 0.090810518 |
| 5 | 5 | 1732 | 424 | 376 | 125 | 0.21709007 | 0.29481132 | 0.077721251 |
| 6 | 6 | 16188 | 6661 | 3334 | 1915 | 0.20595503 | 0.28749437 | 0.081539342 |
| 7 | 7 | 13198 | 4339 | 2474 | 1345 | 0.18745264 | 0.30997926 | 0.122526614 |
| 8 | 8 | 63498 | 63854 | 15465 | 19917 | 0.24355098 | 0.31191468 | 0.068363702 |
| 9 | 9 | 6 | 11 | 1 | 2 | 0.16666667 | 0.18181818 | 0.015151515 |
| 10 | 10 | 54 | 135 | 19 | 43 | 0.35185185 | 0.31851852 | -0.033333333 |

where the columns represent the number of rides per neighborhood, the number of rides during "rush hour" per neighborhood, and the probability in both 2014 and 2019. Then, for each neighborhood, we are testing

$$H_0 : p_{2014} = p_{2019},$$

so it makes sense to use a pooled $z$-test. We first use BH method with $\alpha = 0.10$ and report the results:

| | n_min | m_tested | n_disc |
|---|---|---|---|
| 1 | 0 | 75 | 18 |
| 2 | 10 | 59 | 20 |
| 3 | 25 | 43 | 19 |
| 4 | 50 | 34 | 18 |
| 5 | 100 | 27 | 16 |
| 6 | 200 | 24 | 15 |
| 7 | 400 | 19 | 13 |

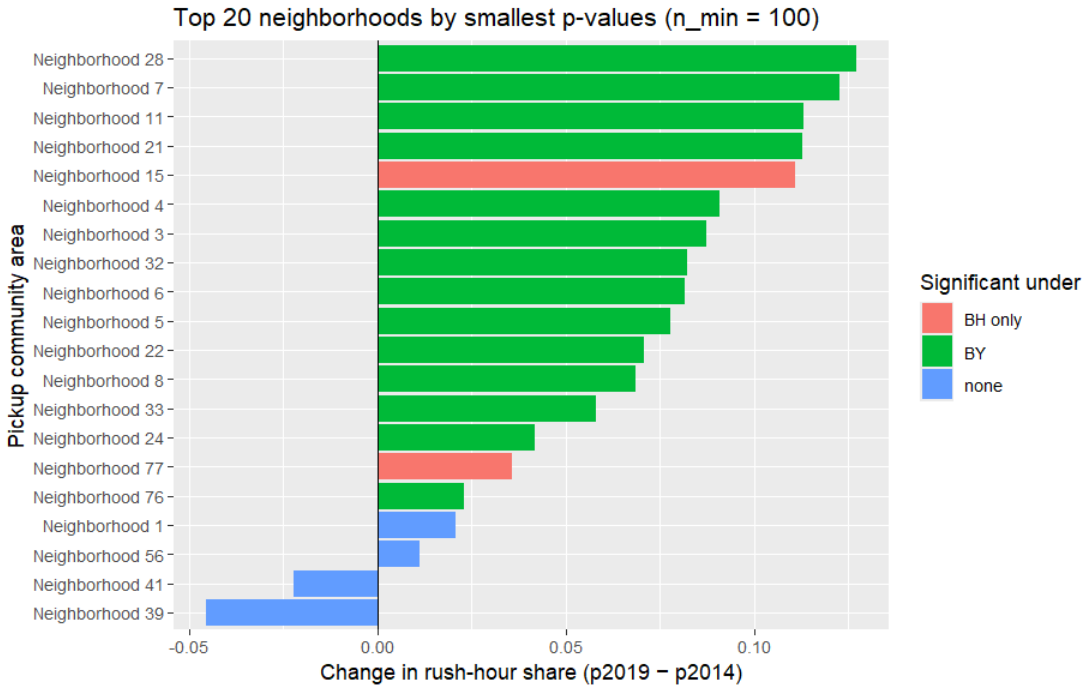



From the above table, we see that 18 neighborhoods experienced a change in rush-hour taxi rides from 2014 to 2019 if $n_{\min} = 100$. We also note that for larger $n_{\min}$, we are starting to only select for larger / busier neighborhoods.

### 4.1.3 Comparison to BY

From the previous section, one of our assumptions was independence between neighborhoods. However, as discussed above, this is not really a realistic assumption that we can make. In this case, using **BY correction may be better**, since BY controls FDR under any arbitrary dependence.

| | n_min | m_tested | n_disc_BH | n_disc_BY |
|---|---|---|---|---|
| 1 | 0 | 75 | 18 | 15 |
| 2 | 10 | 59 | 20 | 15 |
| 3 | 25 | 43 | 19 | 15 |
| 4 | 50 | 34 | 18 | 15 |
| 5 | 100 | 27 | 16 | 14 |
| 6 | 200 | 24 | 15 | 14 |
| 7 | 400 | 19 | 13 | 12 |




The above confirms that BY is strictly more conservative than BH. In addition, this suggests that the neighborhoods in green have strong signals, since they are discoveries under both BY and BH. Furthermore, since the top 20 neighborhoods account for $\approx 97\%$ of the total taxi volume, this result is quite significant, since if we control $n_{\min}$ at 400, more than half of the neighborhoods are reported as a discovery. Therefore, there is a noticeable increase of

## 4.2 Family 2: Neighborhood × Time-Bin Tests

For this test, we clearly have even more degrees of multiplicity compared to the previous, up to 5 times the total number of neighborhoods tested. We make the same assumptions as those from 4.1.1, and use a pooled $z$-test before implementing both BH and BY correction methods.

| | n_min | m_tested | n_disc_BH | n_disc_BY |
|---|---|---|---|---|
| 1 | 0 | 327 | 105 | 74 |
| 2 | 10 | 165 | 96 | 81 |
| 3 | 25 | 123 | 82 | 75 |
| 4 | 50 | 104 | 74 | 68 |
| 5 | 100 | 76 | 61 | 58 |
| 6 | 200 | 55 | 48 | 47 |
| 7 | 400 | 44 | 38 | 37 |

Compared to section 4.1, the importance of $n_{\min}$ is much more apparent. The number of discoveries decreases from 105 to 38 if we use BH, and 74 to 37 if we use BY. This sharp contraction confirms that many of the smallest-cell discoveries are variance-driven rather than signal-driven. From our exploratory data analysis, we know that 20 neighborhoods account for close to 97% of the total taxi volume. In this setting, that means that we should test around $20 \cdot 5 = 100$ total neighborhoods and times, and it should be less given that some periods of the day do not have a lot of taxi volume. Therefore, the number of discoveries if we do not impose a $n_{\min}$ is not really informative, since it puts too much weight on smaller districts that can have huge swings in taxi volume.

In addition, we note that BY compared to BH results in much more conservative results, which suggests

that there is indeed greater dependence when we add time as a differentiator. This is because if we know the total volume of taxi rides in a neighborhood and the volume of taxi rides for four of the five time bins defined above, then the fifth time bin is already known.

## 4.3 Comparison Between Family 1 & Family 2

The effects of greater multiplicity are clearly shown from the two tests above. The sharp decrease in number of discoveries as $n_{\min}$ in creases in section 4.2 suggests that the variance of the $z$-statistic was much larger than the $z$-statistic from section 4.1. In contrast, since section 4.1 only has 1 test per neighborhood, the multiplicity corrections (either BH or BY) do not severely erode power.

In addition, the conclusions that we can make from these two sections are also distinct. From section 4.1, we can infer that there are broad, across-the-board changes in rider behavior during rush hour, whereas in section 4.2, we can detect patterns that cancel out at a higher level. For instance, if a neighborhood had greater morning rides but fewer evening rides that nearly canceled each other out, that neighborhood would not show up as a discovery in section 4.1, but would be highlighted in section 4.2.

## 4.4 Multiplicity Reduction via $n_{\min}$ Thresholding

In sections 4.1 and 4.2, we noticed that increasing $n_{\min}$ decreases the number of discoveries. More formally, since we are running a pooled test, the $z$-statistic for some neighborhood is

$$Z = \frac{\hat{p}_{2019} - \hat{p}_{2014}}{\sqrt{\hat{p}(1-\hat{p}) \cdot (\frac{1}{n_{2019}} + \frac{1}{n_{2014}})}}.$$

For small $n_{\min}$, the $z$-statistic potentially can get large due to the $1/n$ in the denominator. Thus, $n_{\min}$ controls the variance of our $p$-values and does multiplicity reduction simultaneously.

### 4.4.1 Bias-Variance Tradeoff

$n_{\min}$ can therefore be viewed as a form of bias-variance tradeoff, since for large $n$, we decrease variance at the risk of increasing bias, since we are only selecting larger neighborhoods. However, for small $n$, there is less bias, but the variance can be quite large. So, some middling ground would provide the best compromise (ie $n = 100$).

# 5 Additional Outcomes

While Sections 3-5 focused on rush-hour usage as the primary behavioral outcome, to understand the full narrative of taxi rides, we now extend the analysis to examine two additional binary outcomes that capture distinct dimensions of rider behavior: (i) cash payment share at the neighborhood level (Family 1) and (ii) tipping propensity among non-cash rides at the neighborhood $\times$ time-bin level (Family 2). These outcomes are structurally different from ride timing, as they are closer to individual/family-level choice and less mechanically tied to macro changes in commute patterns. The goal here is not merely to test additional hypotheses, but to assess whether the structural shifts observed in ride timing are accompanied by economically meaningful changes in payment behavior.

All tests in this section use two-sample tests of proportions at the neighborhood level:

$$H_{0a} : p_{a,2019} = p_{a,2014} \qquad \text{vs.} \qquad H_{1a} : p_{a,2019} \neq p_{a,2014}$$

where $p_{a,y} = \mathbb{P}[Z = 1 \mid \text{pickup area } = a, \text{year } = y]$. P-values are adjusted using Bonferroni, Holm, BH, BY and Storey corrections as in Section 4. Throughout, we control the target error rate at $\alpha = 0.10$. Note that we continue to restrict to neighborhoods with at least $n_{min}$ rides per year as defaulted and defined earlier. For neighborhood $a$, let

$$\hat{p}_{a,y} = \frac{k_{a,y}}{n_{a,y}}, \qquad \hat{\Delta}_a = \hat{p}_{a,2019} - \hat{p}_{a,2014.}$$

## 5.1 Cash Payment Proportion (F1)

We first examine whether the proportion of rides paid in cash changed between 2014 and 2019 within each neighborhood. Particularly, the hypotheses we are testing are:

$$H_{0a} : p^{cash}_{a,2019} = p^{cash}_{a,2014} \qquad \text{vs.} \qquad H_{1a} : p^{cash}_{a,2019} \neq p^{cash}_{a,2014}$$

where $p^{cash}_{a,y} = \mathbb{P}[cash = 1 \mid \text{pickup area } = a, \text{year } = y]$. After restriction, there are 34 hypotheses being tested in this subsection. Testing is conducted via a two-sample test of proportions using the pooled standard error, namely

$$Z_a = \frac{\hat{p}_{a,2019} - \hat{p}_{a,2014}}{\sqrt{\hat{p}_{a,pool}(1 - \hat{p}_{a,pool})}(\frac{1}{n_{a,2019}} + \frac{1}{n_{a,2014}})}$$

with $\hat{p}_{pool} = \frac{k_{a,2019}+k_{a,2014}}{n_{a,2019}+n_{a,2014}}$.

Figure 5 showcases the estimated effects $\hat{\Delta}_a$ across neighborhoods. The pattern is strikingly asymmetric: nearly all neighborhoods exhibit a negative effect / decline in cash share between 2014 and 2019. The largest decreases exceed 20 percentage points (e.g. area 38 shows approximately -0.22, area 28 shows approximately -0.20), while the median effect across tested neighborhoods is strongly negative. Only a small number of neighborhoods show slight increases (all below +0.03). Importantly, these changes of our focus are proportion instead of volume changes. Thus, this pattern reflects a shift in payment modality conditional on taxi usage, not simply changes in demand composition. The magnitude of these effects, on the order of 10 to 20 percentage points in many neighborhoods, is economically large relative to the baseline cash rate. Thus, this is not a marginal shift but a citywide behavioral reallocation away from cash.

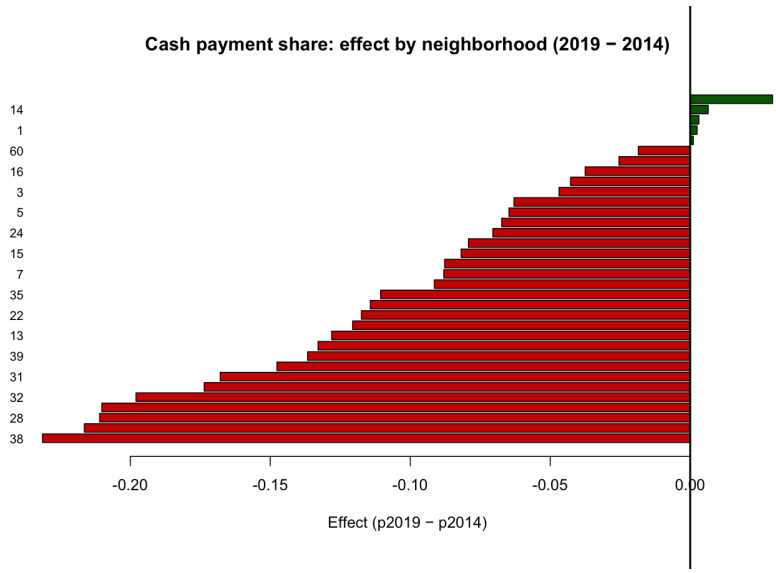



Figure 5: Figure 5.1A: Cash Payment Share Net Change from 2014 - 2019 by Neighborhood

We also assess the histogram of unadjusted p-values for Cash Payment Proportion change in 6. Under the complete global null (no change in any neighborhood), the p-values should be approximately Uniform$(0, 1)$, implying an expected frequency of $m/20$ per bin for each of the 20 bins. Instead, the histogram shows a strong concentration near zero (large spike in first bin), sparse mass in intermediate bins, and extreme left-skewness, with a few large p-values near 1. This deviation from uniformity indicates that the global null is decisively false as a large fraction of false nulls occur. If all nulls were true, $\mathbb{E}[\text{count in each bin}] = 34/20 = 1.7$, but the first bin visibly exceeds this expectation by an order of magnitude. This visually confirms the shift away from cash is statistically systematic rather than noise-drive.

Formally, the empirical distribution of p-values satisfies $\frac{1}{m}\sum_{a=1}^{m} \mathbb{1}\{p_a \leq t\} \gg t$ for small $t$, which implies strong signals under any FDR-based correction.

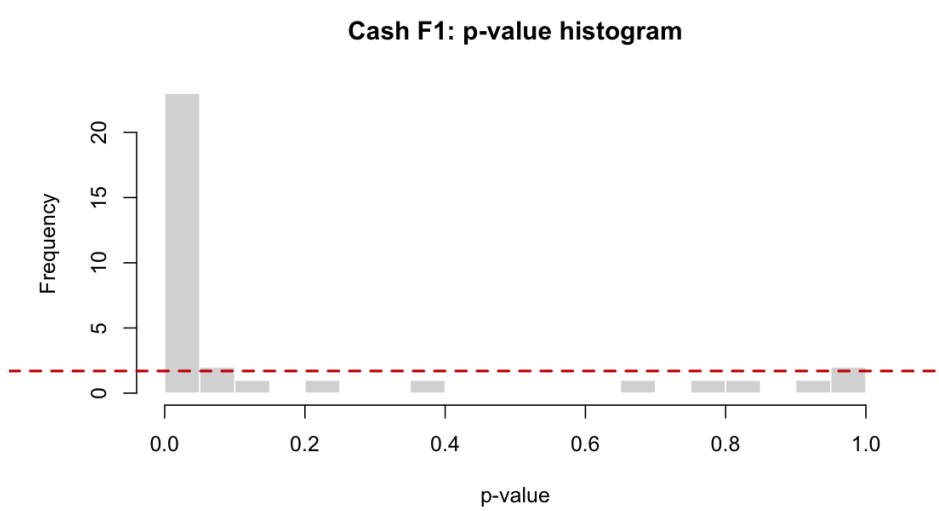



Figure 6: Figure 5.1B: Unadjusted Cash Payment Share Change (Family 1) p-value Histogram

**Multiplicity-Adjusted Discoveries:** After applying multiple testing corrections at $\alpha = 0.10$, we observe the following:

Table 1: Multiplicity-adjusted discoveries for cash payment share (Family 1), $\alpha = 0.10$, $m = 34$ hypotheses.

| Correction Method | Number of Discoveries |
|---|---|
| Bonferroni | 19 |
| Holm | 21 |
| BH | 24 |
| BY | 23 |
| Storey | 26 |

Several features are notable:

- Even the strict Bonferroni correction (FWER control) rejects 19 out of 34 hypotheses.
- FDR-based procedures reject roughly $70 - 75\%$ of tested neighborhoods.
- Storey's adaptive procedure yields the largest number of discoveries, consistent with an estimated $\pi_0 < 1$.

The fact that Bonferroni, controlling $\mathbb{P}[\text{any false rejection}] \leq \alpha$, still rejects more than half of the hypotheses indicates that the effect is not marginal or fragile. We can thus confidently move onto interpretation of our results.

Statistically, the evidence for a reduction in cash payment usage between 2014 and 2019 is overwhelming, consisting of large negative effect sizes, strong deviation of p-values from uniformity, and high discovery counts under even strict control. Substantively, this confirms a structural payment shift rather than localized or idiosyncratic changes. The consistency across neighborhoods suggests a citywide change in payment behavior, likely driven by increased card and technological app-based payment penetration.

We highlight that this outcome differs materially from rush-hour share in Section 4. When evaluated at the same multiplicity level (m = 34 hypotheses), the cash outcome yields 19 Bonferroni rejections and 24 BH rejections, compared to 15 and 18 respectively for rush-hour share. Moreover, cash effect sizes are uniformly negative and often exceed 10–20 percentage points, whereas rush-hour shifts were smaller in magnitude and heterogeneous in sign. Although multiplicity correction changes the exact number of discoveries, the qualitative conclusion of a broad citywide decline in cash usage remains robust even under FWER control. This makes the cash outcome a useful benchmark for the rest of our framework as it represents a high-signal regime in which inferential conclusions are comparatively stable across correction methods.

## 5.2 Tipping Propensity Among Non-Cash (F2)

We next analyze tipping propensity among non-cash rides at the finer neighborhood $\times$ time-bin level (Family 2). Restricting to non-cash rides eliminates the measurement error induced by unrecorded cash tips and allows us to define the binary outcome $Z_{\text{tip}} = \mathbb{1}\{\text{tip} > 0\}$.

For each neighborhood $a$ and time bin $b$, we test the hypotheses

$$H_{0,ab}: \; p_{ab,2019} = p_{ab,2014} \qquad \text{vs.} \qquad H_{1,ab}: \; p_{ab,2019} \neq p_{ab,2014},$$

where $p_{ab,y} = \mathbb{P}(Z_{\text{tip}} = 1 \mid a, b, \text{year} = y)$. Again, the test statistic is the standard two-sample proportion $z$-statistic,

$$Z_{ab} = \frac{\hat{p}_{ab,2019} - \hat{p}_{ab,2014}}{\sqrt{\hat{p}_{ab}(1-\hat{p}_{ab})\left(\frac{1}{n_{ab,2019}} + \frac{1}{n_{ab,2014}}\right)}},$$

with $\hat{p}_{ab}$ being the pooled estimator under $H_0$.

After enforcing the $n_{\min}$ threshold in both years, this subsection yields $m = 61$ hypotheses. The larger multiplicity relative to Section 5.1 arises because Family 2 tests each neighborhood across up to five time bins, whereas Family 1 tested only one hypothesis per neighborhood.

**Multiplicity-Adjusted Discoveries:** After applying multiple testing corrections at $\alpha = 0.10$, we observe the following:

Table 2: Multiplicity-adjusted discoveries for non-cash tip propensity (Family 2), $\alpha = 0.10$, $m = 61$ hypotheses.

| Correction Method | Number of Discoveries |
|---|---|
| Bonferroni | 19 |
| Holm | 20 |
| BH | 26 |
| BY | 21 |
| Storey | 44 |

Even under the stringent Bonferroni control of the FWER, 19 neighborhood $\times$ time-bin cells remain significant. Under BH control of the false discovery rate, 26 cells are rejected, while the adaptive Storey procedure yields 44 discoveries, reflecting the strong left-skew visible in the p-value distribution. As seen in Figure 7 the histogram of raw p-values, the first bin near zero is heavily inflated relative to the Uniform $(0, 1)$ reference line, while the remainder of the distribution is relatively sparse. This pattern would be inconsistent with the global null, indicating that a non-trivial fraction of the 61 hypotheses correspond to year-over-year changes in tipping propensity and providing visual confirmation that many intersection nulls are false.

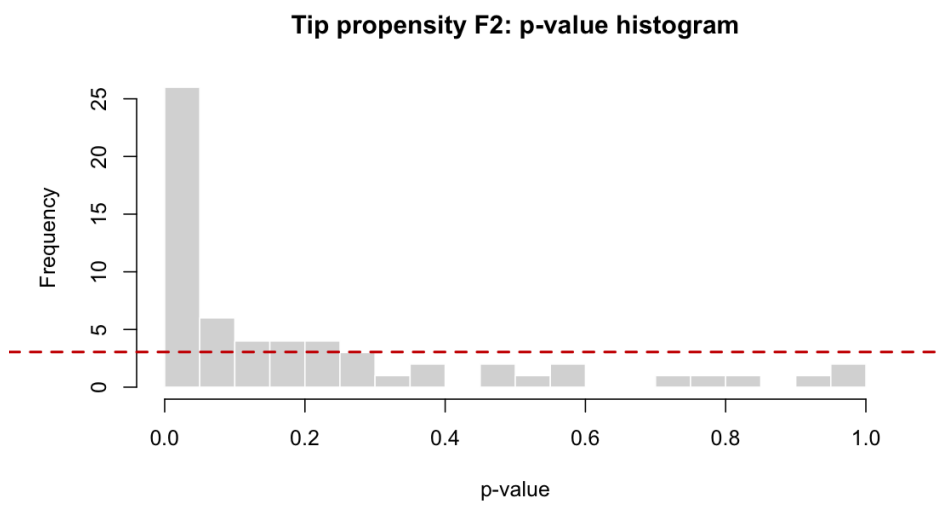



Figure 7: Figure 5.2A: Non-cash Tip Propensity (Family 2) p-value Histogram

We also take a closer look at the spatial structure of effects. Figure 8 displays the effect sizes $\hat{p}_{ab,2019} - \hat{p}_{ab,2014}$ across neighborhoods and time-bins. The heatmap reveals that the strongest effects concentrate in the midday bin, where several neighborhoods exhibit substantial negative shifts in tipping propensity (deep red cells), indicating a decline in tipping propensity among non-cash riders in those periods. Late-night effects are comparatively smaller and less consistent across space.

Relative to cash share where nearly all effects were negative and spatially consistent, tipping propensity exhibits both heterogeneity across time bins and stronger localization. However, the number of BH discoveries (26 out of 61) indicates that the signal remains substantial even after accounting for multiplicity, which justifies the use of the higher-resolution Family 2 design. This structure is statistically meaningful for our investigation. Because the multiplicity burden applies across all 61 cells, concentrated effects in specific bins generate clusters of small p-values, which explains why BH and Storey yield substantially more discoveries than Bonferroni. Nevertheless, the persistence of 19 Bonferroni rejections indicates that the tipping signal is not an artifact of multiplicity or low-powered cells.

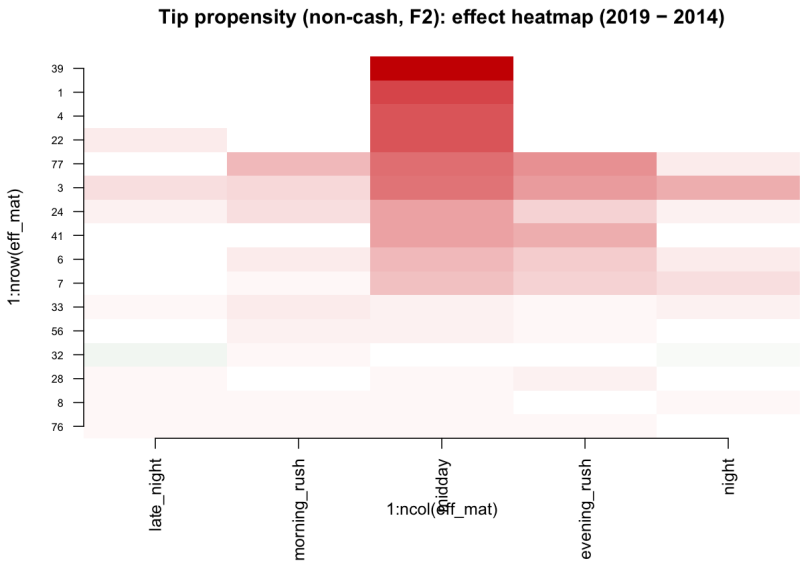



Figure 8: Figure 5.2B: Non-cash Tip Propensity (Family 2) Heatmap

## 5.3 Joint multiplicity across outcomes

Thus far, each outcome (rush-hour share, cash payment share, and tip propensity) has been analyzed within its own hypothesis family. We now implement a more exact inferential framework by pooling all hypotheses into a single macro family and applying multiplicity correction across outcomes.

Specifically, we combine:

- 34 neighborhood-level hypotheses for cash payment share (F1),
- 34 neighborhood-level hypotheses for rush-hour share (F1),
- 61 neighborhood $\times$ time-bin hypotheses for tip propensity (F2),

yielding a total of

$$m = 34 + 34 + 61 = 129$$

simultaneous tests.

We then apply the same correction procedures as before (Bonferroni, Holm, BH, BY, and Storey) at $\alpha = 0.10$. The resulting discovery counts are shown in Table 3.

Table 3: Joint-family discoveries across all outcomes ($m = 129$, $\alpha = 0.10$)

| Method | Bonferroni | Holm | BH | BY | Storey |
|---|---|---|---|---|---|
| Discoveries | 47 | 49 | 69 | 57 | 86 |

Figure 9 below visualizes these raw discovery counts. The ordering is consistent with theoretical guarantees: Bonferroni and Holm (strong FWER control) are most conservative; BH controls FDR under independence; BY adjusts for arbitrary dependence and is therefore more conservative than BH; Storey's adaptive estimator of $\pi_0$ yields the largest number of rejections.

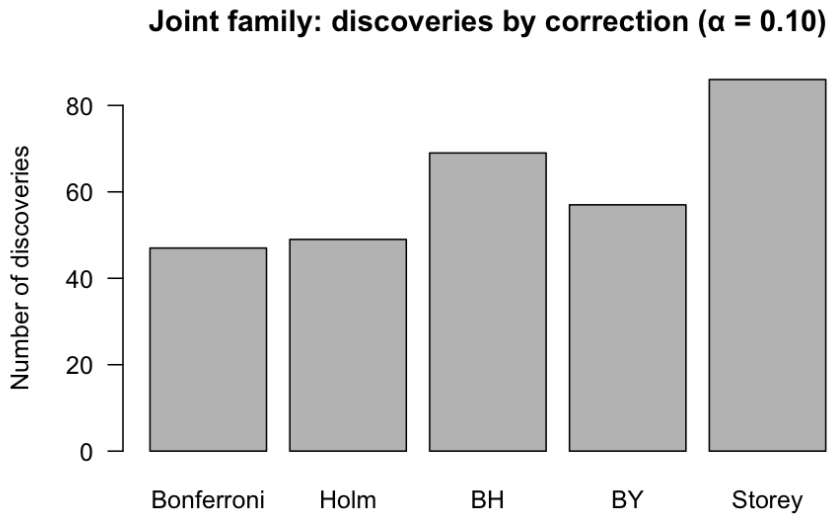



Figure 9: Figure 5.3: Joint Discoveries Bar Plot

We notice that the joint-family correction reduces discoveries relative to analyzing each outcome separately. For example, under BH procedure: BH discoveries (cash F1) = 24,

BH discoveries (tip F2) = 26, whereas pooling all outcomes yields BH discoveries (joint) = 69, which is strictly smaller than the sum of separate-family BH discoveries due to the increased multiplicity burden. Most importantly, the joint framework helps us answer a stronger scientific question: Across all behavioral dimensions studied, how many statistically reliable changes can we assert while controlling error at the family level?

The joint-family analysis clarifies the relative strength and structure of the three behavioral dimensions: rush-hour usage, cash payment share, and tipping propensity. When analyzed separately in Sections 4 and 5.1–5.2, each outcome produced statistically significant discoveries. However, when pooled into a single family of $m = 129$ hypotheses, the outcomes do not contribute equally to the Bonferroni-robust core.

Cash payment share exhibits the strongest and most spatially consistent signal. A large fraction of its neighborhood level hypotheses survive even with the joint Bonferroni threshold,

$$p_i \leq \frac{0.10}{129} \approx 7.75 \times 10^{-4},$$

indicating extremely small p-values and large effect magnitudes. The shift away from cash therefore represents the most statistically dominant behavioral change between 2014 and 2019.

Tipping propensity, analyzed at the neighborhood $\times$ time-bin level, produces substantial but more structured signal. Many cells survive FDR control under BH and Storey, but fewer survive Bonferroni correction in the joint family. This indicates that tipping changes are real but more concentrated in particular time bins rather than uniformly across neighborhoods.

Rush-hour share is comparatively modest in the extreme tail of the p-value distribution. While it shows meaningful changes under FDR control and at the neighborhood level, fewer of its hypotheses remain in the Bonferroni-robust core once all outcomes compete for the same error bandwidth. This suggests that rush-hour effects are moderate in magnitude relative to the payment-based outcomes.

The joint-family results therefore establish a hierarchy of behavioral signal strength: Cash shift $>$ Tipping shift $>$ Rush-hour shift in terms of correction-insensitive statistical evidence.

Numerically speaking, the persistence of 47 Bonferroni discoveries at $m = 129$ shows that a substantial subset of effects are not sensitive to outcome multiplicity control. These correspond primarily to payment-

related changes. The additional discoveries under BH (69 total) and Storey (86 total) reflect moderate-strength effects that become detectable when controlling the FDR rather than the FWER.

Thus, the joint analysis does not merely confirm that the 2014-2019 period saw widespread and systematic changes across multiple dimensions of taxi rider behavior: it differentiates which dimensions changed most strongly.

## 5.4 The Effect of Methodological Choices

Across Sections 4 and 5, the central object of study was not merely whether behavioral change occurred, but how inference depends on methodological design. Several choices materially altered the size and composition of the rejection set.

First, the definition of the hypothesis family determined the multiplicity burden. Moving from neighborhood-level tests (Family 1) to neighborhood $\times$ time-bin tests (Family 2) increased $m$ and reduced power for marginal effects, but revealed structured patterns that were invisible at aggregate resolution. Likewise, pooling all outcomes into a joint family ($m = 129$) imposed a stricter global error budget. Thus, increasing multiplicity did not overturn conclusions; it sharpened which effects are strongest and which depend on the chosen error criterion.

Second, the choice of error metric (FWER vs FDR) directly influenced the boundary between "strong" and "moderate" signal. Bonferroni and Holm isolated effects with extremely small p-values, while BH and Storey expanded the rejection set to include moderate but consistent shifts. The gap between BH and BY further revealed dependence across hypotheses, indicating spatial and cross-outcome correlation. In this dataset, the substantive conclusions are robust to switching between FWER and FDR control for the strongest effects, but moderate-strength findings are sensitive to this choice.

Thirdly, outcome construction itself mattered. By defining binary indicators (rush-hour share, cash share, tip propensity) rather than counts or dollar amounts, we insulated inference from overall demand shifts and price-level changes. Restricting tipping to non-cash rides prevented structural missingness from contaminating the null distribution. These design choices ensured that detected effects reflect behavioral composition rather than scale effects.

Taken together, the analysis demonstrates that methodological choices do not simply scale the number of discoveries. Higher resolution increases sensitivity to structured heterogeneity. Stricter multiplicity control isolates the extreme tail of the signal distribution. Adaptive procedures reallocate power when the empirical null proportion is low. The robustness of the critical findings across these pathways strengthens the substantive conclusion, while the variability in marginal discoveries highlights the importance of transparently reporting methodological dependence.

# 6 Validation & Diagnostics

## 6.1 Placebo Test

A credible multiple testing pipeline must demonstrate that its discovery rate is controlled under the null. Our initial approach – testing "fake" rush-hour windows (e.g., hours 12-14 and 21-23) against the same 2014-vs-2019 comparison – produced an unacceptable number of rejections. This failure is expected in hindsight: as documented in Section 3.2, the entire hourly distribution of taxi rides shifted between 2014 and 2019, so any arbitrary set of hours exhibits a statistically significant change in share between years. A valid placebo must therefore construct a comparison in which no true effect exists by construction, while preserving the same family of hypotheses, test statistics, correction methods, and approximate sample sizes as the real analysis. We implement two complementary approaches that satisfy these requirements.

### 6.1.1 Placebo A: Within-Year Random Split

We take all rides from a single year (either 2014 or 2019), randomly assign each ride to one of two arms (labeled "A" and "B"), and apply the identical F1 z-test pipeline with BH correction. Because both arms are independent random draws from the same population, every null hypothesis is exactly true. Any rejection is a genuine false positive. We set $n_{min} = 25$ (half the main analysis threshold) to maintain a comparable number of testable hypotheses given the halved per-arm sample size. We repeat this procedure across 50 independent random splits to estimate the distribution of false discovery counts.

Across 50 replications, all 20 neighborhoods were testable in every split. The mean number of BH discoveries was 0.20, with 43 of 50 replications producing zero false discoveries and a maximum of 3. The pooled p-value distribution across all replications is approximately uniform (right panel), confirming that the z-test is well-calibrated under the null. Results for the within-2019 split were comparable (mean BH discoveries: 0.16).

The expected number of false discoveries under a perfectly calibrated BH procedure applied to $m = 20$ independent null $p$-values at $\alpha = 0.10$ is at most $\alpha \times m = 2.0$. Our observed rate of 0.20 is an order of magnitude below this bound, indicating that the procedure is conservative here likely because the discrete nature of the proportion $z$-test produces slightly conservative $p$-values. This conservatism is reassuring.

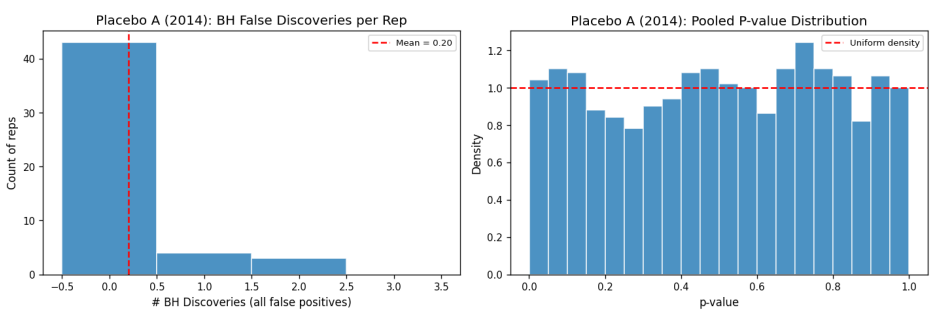



Figure 10: Placebo A (within-2014 split). Left: Distribution of BH false discoveries across 50 random splits. Right: Pooled p-value histogram showing approximate uniformity.

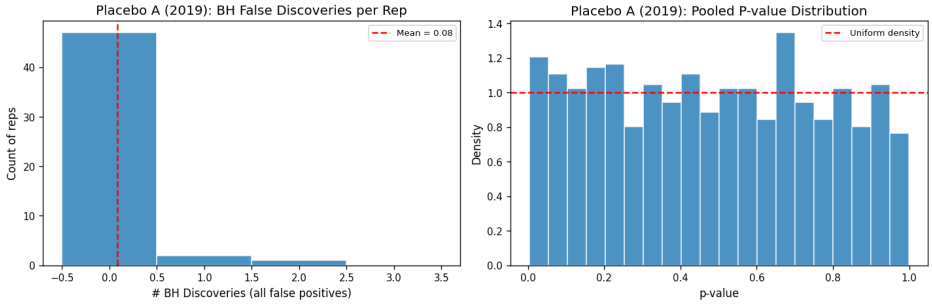



Figure 11: Placebo A (within-2019 split). Left: Distribution of BH false discoveries across 50 random splits. Right: Pooled p-value histogram showing approximate uniformity.

### 6.1.2 Placebo B: Year-Label Permutation Null

We pool all rides from both 2014 and 2019, randomly shuffle the year labels globally (breaking the true temporal structure), and run the full F1 $z$-test pipeline with BH correction. This tests the global null $H_0$: year labels are uninformative for all neighborhoods simultaneously. This is repeated 50 times to construct the null distribution of the number of discoveries. Unlike Placebo A, this approach preserves the original sample sizes and the exact same $n_{\min}$ threshold as the real analysis, providing a direct null benchmark for comparison.

Under the permutation null, the mean number of BH discoveries was 0.08, with a maximum of 1 across 50 replications. In contrast, the real 2014-vs-2019 comparison produced 7 BH discoveries. No permutation replicate came close to the real count: the pipeline-level permutation p-value is $\hat{p} = (1+0)/(50+1) = 0.0196$. The pooled null p-value histogram is again approximately uniform (right panel).

The real discovery count of 7 lies far outside the permutation null distribution (left panel). This provides strong evidence that the detected rush-hour shifts are not artifacts of the statistical pipeline or residual dependence structure in the data, but reflect genuine behavioral differences between the two years.

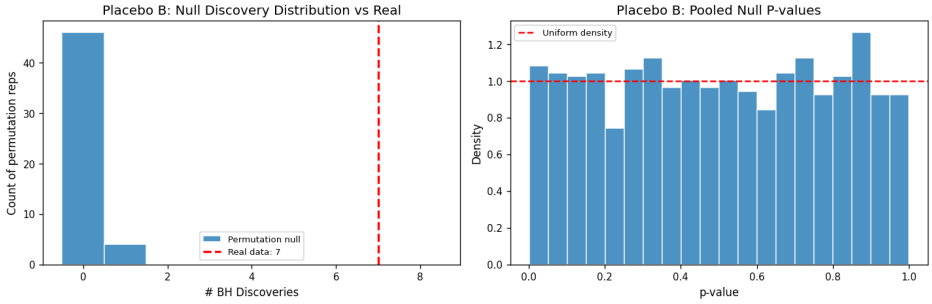



Figure 12: Placebo B (year-label permutation null). Left: Null distribution of BH discoveries (blue) versus real discovery count (red dashed line at 7). Right: Pooled null p-value distribution.

### 6.1.3 Summary

Both placebo approaches confirm that our testing pipeline controls false discoveries at or below the nominal level. Within-year split validates the calibration of the test statistic and correction procedure in isolation; the year-label permutation validates the full pipeline against the null of no year effect. We repeat both placebo procedures for the cash payment outcome. The within-2014 split yields a mean of 0.30 BH false discoveries per replication; the within-2019 split yields 0.10; and the year-label permutation null yields 0.20. The real analysis produces 16 BH discoveries. The separation is even more dramatic than for rush-hour timing, confirming that the widespread cash decline documented in the EDA is not a pipeline artifact. The large gap between null and real discovery counts (7 vs. $\leq$1 for rush-hour; 16 vs. $\leq$1 for cash) provides strong evidence that our findings reflect genuine behavioral shifts.

## 6.2 Stability and Reproducibility

A discovery that disappears when the data are mildly perturbed is difficult to trust. To assess the robustness of our findings, we apply a subsample stability analysis: we independently draw 50% of the 2014 rides and 50% of the 2019 rides (without replacement), run the identical F1 $z$-test pipeline with BH correction at $\alpha = 0.10$, and record the discovery set. We repeat this procedure 30 times and measure three quantities: (i) the Jaccard index between each subsample's discovery set and the full-sample discoveries, (ii) the distribution of the number of discoveries across subsamples, and (iii) the per-neighborhood recovery rate (how frequently each full-sample discovery reappears in subsamples).

### 6.2.1 Rush-Hour Timing (F1)

The full-sample analysis identified 7 neighborhoods with significant changes in rush-hour share under BH: areas 8, 16, 28, 32, 33, 41, and 76. Across 30 half-sample replications, the mean number of discoveries was $5.2 \pm 1.6$, with a mean Jaccard overlap of $0.630 \pm 0.145$ against the full-sample set.

The per-neighborhood recovery rates (left panel) reveal a stratification of the 7 discoveries into a stable core and a fragile periphery. Areas 28 and 32 were recovered in 100% of subsamples; area 8 in 97%; and area 41 in 83%. These four neighborhoods represent robust signals that survive substantial data reduction. In contrast, areas 16 (23%), 76 (33%), and 33 (37%) are marginal discoveries whose inclusion in the rejection set depends sensitively on sample composition. This stratification is expected: neighborhoods with larger effect sizes and larger sample sizes are recovered more reliably, while those near the BH rejection boundary are inherently fragile.

The left panel also reveals that a handful of neighborhoods not in the full-sample discovery set (shown in red) appear occasionally in subsamples. These are neighborhoods whose $p$-values hover just above the BH threshold in the full sample but occasionally cross it when the random subsample happens to produce a slightly more extreme test statistic. Their low recovery rates (below 15%) confirm that they are not robust findings.

The discovery count distribution (middle panel) clusters between 4 and 7, with the mode at 5 and the full-sample count of 7 at the upper end. That subsamples typically recover fewer discoveries than the full sample is expected given the 50% reduction in sample size, which reduces both the power of individual tests and the effective signal available to the BH step-up procedure.

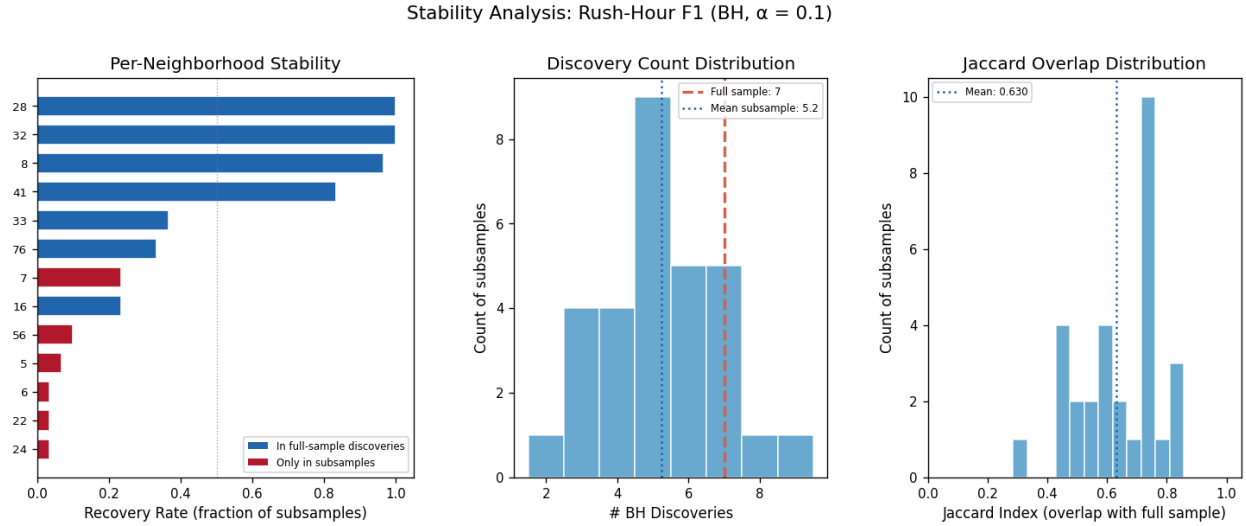



Figure 13: Stability analysis for rush-hour F1 (BH, $\alpha = 0.10$). *Left:* Per-neighborhood recovery rate across 30 half-sample replications. Blue bars indicate neighborhoods in the full-sample discovery set; red bars indicate neighborhoods discovered only in subsamples. The vertical dotted line marks 50% recovery. *Middle:* Distribution of subsample discovery counts; the red dashed line marks the full-sample count (7). *Right:* Distribution of Jaccard overlap with the full-sample discovery set.

### 6.2.2 Cash Payment (F1)

For the cash outcome, the full-sample analysis identified 16 neighborhoods with significant declines in cash payment share. The stability analysis shows markedly higher robustness: a mean Jaccard overlap of $0.897 \pm 0.064$ with $14.8 \pm 1.0$ mean discoveries per subsample (Figure 6.4). Of the 16 full-sample discoveries, 11 were recovered in every single subsample (100% recovery rate), and only 2 neighborhoods (areas 2 and 77) fell below 70%. The near-perfect stability is consistent with the large, pervasive effect sizes observed in the EDA.

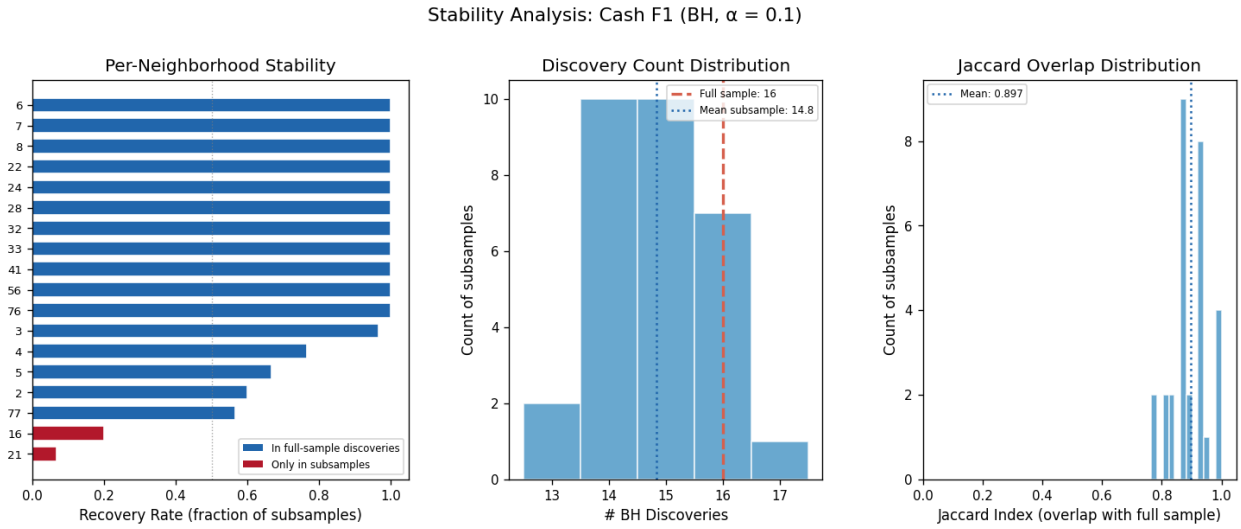



Figure 14: Stability analysis for cash F1.

The stability analysis reveals that our discoveries fall into two regimes. For outcomes with large, widespread effects (cash payment), the discovery set is highly stable: Jaccard overlap of 0.90 and near-perfect per-neighborhood recovery. For outcomes with more heterogeneous, moderate effects (rush-hour timing), a core of 4 neighborhoods is robust (recovery $\geq 83\%$), while 3 peripheral neighborhoods are sensitive to sample composition (recovery $\leq 37\%$). This distinction quantifies which findings should be treated as firmly established versus which require the full sample to detect and should be interpreted with greater caution.

## 6.3 P-value Diagnostics

The distribution of $p$-values across a hypothesis family provides a diagnostic for both the presence of signal and the calibration of the testing procedure. Under a global null, $p$-values should follow a Uniform$(0, 1)$ distribution. Departures from uniformity indicate either the presence of real effects (left skew, excess of small $p$-values) or miscalibration of the test (systematic inflation or deflation). We examine $p$-value histograms and QQ plots for all six analysis pipelines, supplemented by the Kolmogorov-Smirnov (KS) test against the uniform distribution as a formal calibration check.

Figure presents the comprehensive diagnostics. For each pipeline, the left panel shows the $p$-value histogram with a uniform reference line, annotated with the KS statistic and the fraction of $p$-values falling below 0.05 and 0.10; the right panel shows the QQ plot against Uniform$(0, 1)$ with 95% pointwise confidence bands derived from Beta order-statistic quantiles.

**Cross-engine consistency for rush-hour F1.** The three $p$-value engines ($z$-test, logistic regression, permutation) produce nearly identical diagnostic profiles: KS statistics of 0.335-0.353, and an identical fraction of 0.40 of $p$-values below 0.10. This convergence is noteworthy because the three engines make different assumptions – the $z$-test assumes a normal approximation and independence, the logistic model adjusts for month and weekend effects, and the permutation test makes no distributional assumptions but requires exchangeability within strata. That all three agree closely indicates that the detected signals are

| Pipeline | $m$ | KS stat | KS $p$ | Frac $< 0.05$ | Frac $< 0.10$ | Interpretation |
|---|---|---|---|---|---|---|
| Rush F1 ($z$-test) | 20 | 0.353 | 0.010 | 0.40 | 0.40 | Strong signal |
| Rush F1 (logistic) | 20 | 0.335 | 0.017 | 0.35 | 0.40 | Strong signal |
| Rush F1 (permutation) | 20 | 0.337 | 0.016 | 0.35 | 0.40 | Strong signal |
| Rush F2 ($z$-test) | 20 | 0.301 | 0.042 | 0.25 | 0.35 | Strong signal |
| Cash F1 ($z$-test) | 20 | 0.791 | 0.000 | 0.80 | 0.80 | Strong signal |
| Tip F2 ($z$-test) | 62 | 0.500 | 0.000 | 0.50 | 0.56 | Strong signal |

Table 4: $P$-value diagnostic summary across all six testing pipelines. Under the global null (no true effects), $p$-values follow Uniform$(0, 1)$, implying KS $p > 0.10$ and Frac $< 0.05 \approx 0.05$. All six families show significant departures, confirming genuine signal.

robust to modeling choices and that the covariate adjustment does not materially alter the conclusions. This consistency also provides indirect evidence that the normal approximation underlying the $z$-test is adequate for these sample sizes. This finding directly addresses pillar 3 of our research question: in this dataset, the choice between parametric and permutation-based calibration does not materially affect the discovery set. The spatial and temporal dependence present in the data, while real, does not inflate the parametric test's false positive rate to a degree detectable by our diagnostics. This is consistent with the large per-cell sample sizes ($n_{\min} = 50$), which make the normal approximation reliable even under moderate dependence.

**Cash payment signal strength:** The cash F1 family shows the most extreme departure from uniformity: 80% of $p$-values fall below 0.05, and the KS statistic of 0.791 is the largest observed. The QQ plot shows virtually all points pressed against the lower boundary of the confidence band. This is consistent with the EDA finding that cash usage declined in 18 of 20 neighborhoods by 5-20 percentage points – the signal is so pervasive that only a few neighborhoods remain consistent with the null.

**Tipping propensity largest family size:** The tip F2 family contains the most hypotheses ($m = 62$) and still shows strong left skew (50% below 0.05). The QQ plot reveals that the departure from uniformity is concentrated among the smallest $p$-values (roughly the bottom tercile), while the remaining $p$-values track the uniform reasonably well. This pattern is consistent with a mixture model in which a substantial fraction of neighborhood-time-bin cells experienced true changes in tipping behavior, while others did not - the scenario where BH is more powerful relative to FWER methods.

**Calibration assessment:** Our placebo tests in Section 6.1 established that the $z$-test produces well-calibrated, approximately uniform $p$-values when no true effect exists. The departures observed in the real data therefore reflect real signal rather than inflated type I error. The QQ confidence bands reinforce this interpretation: in the placebo analysis, observed $p$-values fall within the bands; in the real analysis they fall systematically below them, a pattern consistent with signal improvement rather than test misspecification.

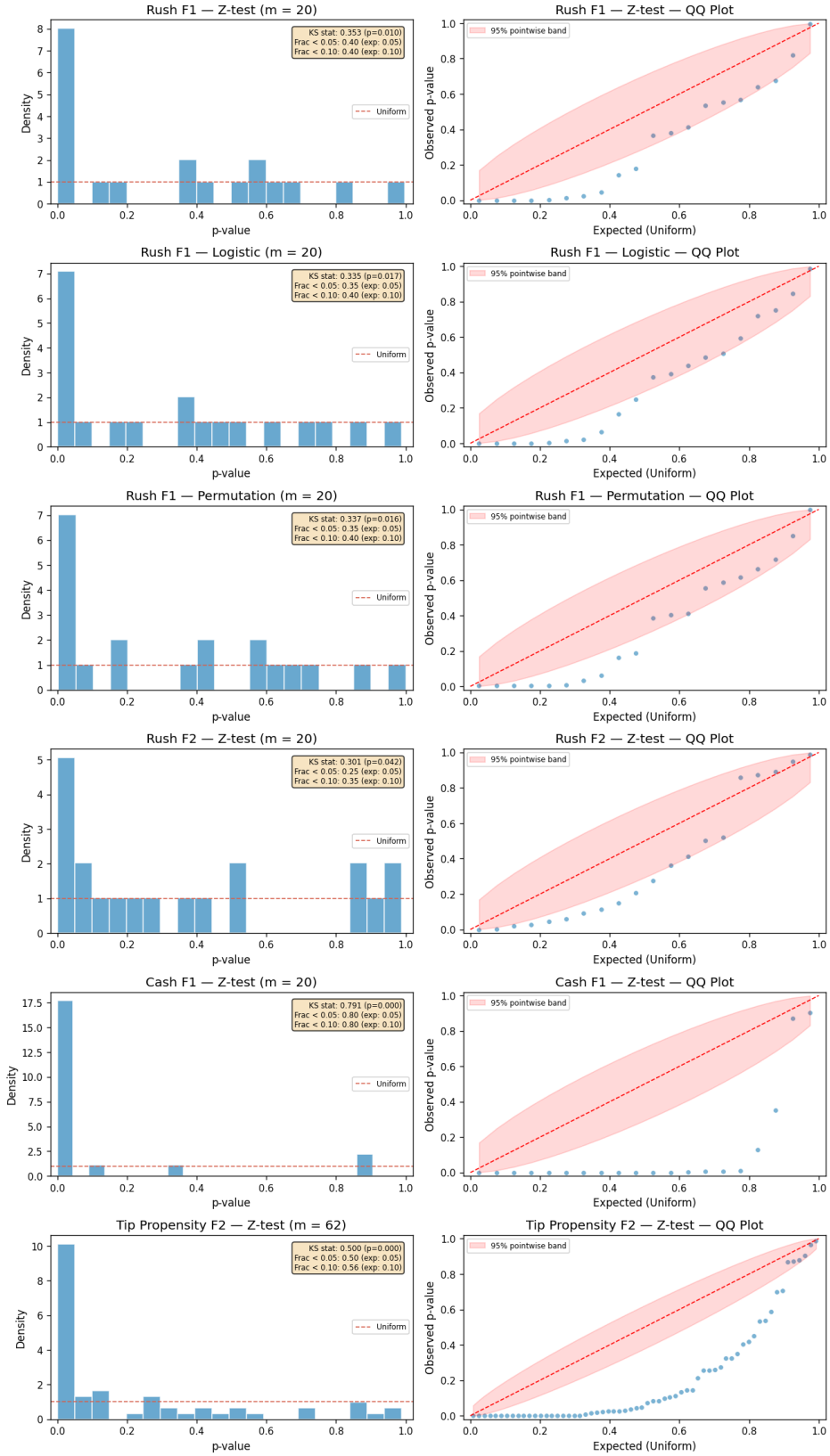



Figure 15: Comprehensive $p$-value diagnostics for all six analysis pipelines. Each row shows the $p$-value histogram (left, with uniform reference and KS annotation) and QQ plot against Uniform$(0, 1)$ (right, with 95% pointwise confidence bands).

#### 6.3.1 Summary

The p-value diagnostics support three conclusions. First, all three outcomes contain real signal and the global null is rejected for every family. Second, the three p-value engines for rush-hour F1 produce consistent diagnostics, confirming that the results are not sensitive to the choice of test statistic or the assumption of independence. Third, the degree of signal enrichment varies across outcomes in a manner aligning with the EDA: cash payment shows the strongest departure from uniformity, followed by tipping propensity (widespread but heterogeneous change), and then rush-hour timing (moderate, spatially varying change). These gradations match the stability results, where cash discoveries were nearly perfectly stable and rush-hour discoveries showed a core-outer structure.

# 7 Discussion & Conclusions

## 7.1 Summary of Findings

We examined neighborhood-level changes in Chicago taxi rider behavior between 2014 and 2019 through a multiple testing framework, focusing on three binary outcomes: rush-hour usage share, cash payment share, and tipping propensity. Each was analyzed under two hypotheses family designs of differing granularity. All three outcomes exhibit statistically detectable changes between 2014 and 2019, but the strength and structure of those changes differ.

Rush-hour share shows moderate shifts with clear spatial heterogeneity. The temporal distribution of taxi rides shifted substantially toward earlier hours, with late-night and morning-rush shares increasing and evening-rush and night shares declining across most neighborhoods. Under BH correction at $\alpha = 0.10$ with $n_{\min} = 100$, Family 1 identified 16 of 27 tested neighborhoods as having significant changes in rush-hour proportion, while the more conservative BY correction identified 14. The effect sizes were moderate (1-8.5 percentage points), and stability analysis revealed a core of 4 robustly significant neighborhoods (28, 32, 8, and 41, with recovery rates $\geq 83\%$) alongside 3 marginal discoveries (16, 76, and 33, with recovery $\leq 37\%$). The heterogeneity in both sign and magnitude of these shifts: most neighborhoods experienced modest declines while areas 41 (Hyde Park) and 16 (Irving Park) showed substantial increases, underscores that aggregate testing would have been insufficient to capture this spatial structure.

Cash payment share produces the strongest and most spatially consistent pattern. Nearly all neighborhoods exhibited negative effects, with magnitudes often exceeding 10-20 percentage points. Even the most conservative correction (Bonferroni, controlling FWER) rejected 19 of 34 hypotheses, while BH rejected 24. The p-value histogram was dramatically left-skewed (80% of p-values below 0.05), and the stability analysis yielded a mean Jaccard overlap of 0.897 with the full-sample discovery set, with 11 of 16 discoveries recovered in every subsample.

Tipping propensity, analyzed at the finer neighborhood $\times$ time-bin level, reveals widespread but heterogeneous change, with 50% of p-values falling below 0.05. The signal, while strong in aggregate, exhibited the mixture structure characteristic of settings where BH-type corrections are most advantageous relative to FWER methods: a substantial fraction of neighborhood - time-bin cells showed genuine behavioral shifts while others did not.

Taken together, these results indicate that the 2014–2019 period involved substantial restructuring of the taxi rider population, with particularly strong changes in payment behavior and more nuanced shifts in temporal usage patterns.

## 7.2 Limitations & Open Questions

A key limitation of this project is that the analysis isolates the taxi market without incorporating rideshare data. Between 2014 and 2019, Uber and Lyft expanded dramatically. If late-night or price-sensitive riders disproportionately migrated to rideshare, the remaining taxi population in 2019 would represent a selected subset of the 2014 population. In that case, our estimates reflect compositional change in the observed taxi rider base rather than pure behavioral change at the individual level.

We also had some attempts that were not fully accomplished due to time constraint. While we explored covariate adjustment through per-neighborhood logistic regressions, we did not implement a fully pooled

model with neighborhood fixed effects and interaction terms. Such a model could test all neighborhoods jointly while directly estimating heterogeneity in year effects. However, translating those joint-model coefficients into a clean multiple testing framework with controlled error rates proved nontrivial. Similar attempts were made to design stronger placebo tests, but that would require resampling or year-label permutation at scale, which becomes computationally expensive when embedded within the full multiplicity framework.

Several methodological questions also emerged during the analysis. First, the choice of $n_{\min}$ materially alters the hypothesis family size. Increasing $n_{\min}$ reduces variance and stabilizes $z$-statistics but also changes the scientific question by excluding low-volume neighborhoods. This raises a difficult tradeoff of whether one should prioritize coverage of all neighborhoods or reliability of inference within well-sampled ones.

Second, dependence across hypotheses remains only partially characterized. Neighborhoods are spatially adjacent, and time bins overlap in behavioral structure. While BY correction protects against arbitrary dependence, it does not model it. A structured approach such as hierarchical modeling could potentially increase power while retaining error control.

In addition, one conceptual curiosity arose when assessing the joint-family analysis: when multiple behavioral dimensions are tested simultaneously, should they be treated as separate scientific families or as components of a single behavioral system? The difference is not merely technical, as it changes the interpretation of "significant change."

On a more macroscopic level, our additional outcomes, especially on tipping, suggests heterogeneity that may interact with payment type and neighborhood income composition—variables not available in the current dataset. Incorporating socioeconomic factors as covariates could clarify whether observed tipping changes reflect behavioral norms, rider composition, or platform-induced changes in payment interface.

## 7.3 Methodological Takeaways & Potential Next Steps

### 7.3.1 Robustness via Outcome Definition and Sample Filtering:

Methodological robustness is effectively established through ex ante design choices rather than post-hoc adjustments. Defining outcomes as binary indicators or proportions mitigates common confounders such as fare inflation and aggregate demand shifts without requiring complex explicit modeling. Concurrently, treating the minimum sample size, $n_{\min}$, as a methodological lever rather than a simple quality filter reduces the multiplicity burden. This data-driven filtering optimizes the bias–variance tradeoff, stabilizing individual test statistics in settings where testable units exceed those with sufficient data for reliable inference.

### 7.3.2 Multi-Facet Validation and Sensitivity Analysis:

Inferential reliability is best assessed through a collection of validation methods rather than a single metric. For internal validity, the combination of placebo tests and stability analysis distinguishes robust core findings from fragile peripheral ones. For model independence, the agreement between parametric (z-test, logistic) and non-parametric (permutation) estimators serves as a critical diagnostic. Convergence across engines confirms that conclusions are not fruits of manipulated modeling assumptions or inflated by spatial-temporal dependence.

### 7.3.3 Conservatism in FDR Control:

The Benjamini-Yekutieli (BY) procedure provides a practical alternative to the Benjamini-Hochberg (BH) correction when dependence structures are uncertain. In this analysis, BY resulted in a negligible reduction in discovery count compared to BH. This suggests that for spatial or panel data, BY offers a cost-effective guarantee of False Discovery Rate (FDR) control under arbitrary dependence, avoiding the excessive conservatism of Bonferroni without risking the potential liberalism of BH. Also, ideally, we can formalize modeling cross-neighborhood dependence and estimate the actual spatial or temporal correlation structure among neighborhoods, but implementing and validating such a structure was beyond the project's scope.

#### 7.3.4 Conclusion:

The analysis confirms statistically significant shifts in Chicago taxi ridership between 2014 and 2019, specifically a widespread transition to earlier ride times and a decline in cash payments. These conclusions are robust to variations in multiple testing corrections, p-value estimators, and sample-size thresholds. However, given the inability to control for rideshare market entry, these findings represents changes in the observed taxi rider population—potentially reflecting compositional shifts in the user base—rather than exclusively individual-level behavioral changes.

# 8 Appendix & Code

## Section 2 Code

```
# Plotting defaults
plt.rcParams.update({
    "figure.dpi": 120,
    "figure.figsize": (8, 4),
    "axes.titlesize": 12,
    "axes.labelsize": 10,
    "font.size": 9,
})

# Reproducibility
RNG_SEED = 27850
rng = np.random.default_rng(RNG_SEED)

#Timezone & Temporal Definitions
LOCAL_TZ = "America/Chicago"

RUSH_HOURS = {7, 8, 9, 16, 17, 18}   # Morning 7-9, Evening 4-6 PM

TIME_BINS = [
    ("late_night",     0,  6),   # 00:00  05  :59
    ("morning_rush",   6, 10),   # 06:00  09  :59
    ("midday",        10, 16),   # 10:00  15  :59
    ("evening_rush", 16, 19),   # 16:00  18  :59
    ("night",         19, 24),   # 19:00  23  :59
]

# Cleaning Thresholds
MAX_TRIP_MILES   = 100.0
MAX_TRIP_SECONDS = 4 * 3600.0   # 4 hours
MAX_TRIP_TOTAL   = 500.0
MAX_EXTRAS       = 100.0
MAX_FARE         = 500.0
MAX_MPH          = 100.0  # Physics plausibility

# Analysis Scope
TOP_K  = 20     # Top neighborhoods by ride count
NMIN   = 50     # Minimum per-year cell size for hypothesis inclusion
ALPHA  = 0.10   # FDR / FWER level
B_PERM = 300    # Permutation replicates

# Assuming df_raw is already loaded as per your previous snippet
df_raw = pd.read_csv("taxidata-1.csv")
df = df_raw.copy()

print(f"Initial Shape: {df.shape}")

# Parse timestamps (UTC -> Local)
df["trip_start_timestamp"] = pd.to_datetime(df["trip_start_timestamp"], utc=True, errors="
df["ts_local"] = df["trip_start_timestamp"].dt.tz_convert(LOCAL_TZ)
```

```
# Filter for relevant years immediately
df["year"] = df["ts_local"].dt.year.astype(int)
df = df.loc[df["year"].isin([2014, 2019])].copy()

# Validity Check (Negatives/Zeros)
# We strictly remove non-positive duration/distance and negative costs
mask_invalid = (
    (df["trip_seconds"] <= 0) |
    (df["trip_miles"] <= 0) |
    (df["fare"] < 0) |
    (df["tips"] < 0) |
    (df["tolls"] < 0) |
    (df["extras"] < 0) |
    (df["trip_total"] < 0)
)
n_invalid = mask_invalid.sum()
df = df.loc[~mask_invalid].copy()

# B. Outlier Capping (Economic & Physical Limits)
mask_outliers = (
    (df["trip_miles"] > MAX_TRIP_MILES) |
    (df["trip_seconds"] > MAX_TRIP_SECONDS) |
    (df["trip_total"] > MAX_TRIP_TOTAL) |
    (df["extras"] > MAX_EXTRAS) |
    (df["fare"] > MAX_FARE)
)
n_outliers = mask_outliers.sum()
df = df.loc[~mask_outliers].copy()

# C. Physics Check (The "Teleportation" Filter)
# Calculate implied speed (MPH)
# This addresses the rubric's "Is data realistic?" requirement
df["mph"] = df["trip_miles"] / (df["trip_seconds"] / 3600.0)
mask_teleport = (df["mph"] > MAX_MPH)
n_teleport = mask_teleport.sum()
df = df.loc[~mask_teleport].copy()

print("\n--- Cleaning Summary ---")
print(f"Removed Invalid (Negatives/Zeros): {n_invalid}")
print(f"Removed Extreme Outliers (Caps):   {n_outliers}")
print(f"Removed Physics Violations (>100mph): {n_teleport}")
print(f"Final Cleaned Shape: {df.shape}")

# Temporal Features
df["month"] = df["ts_local"].dt.month
df["dow"] = df["ts_local"].dt.dayofweek
df["hour"] = df["ts_local"].dt.hour
df["is_weekend"] = (df["dow"] >= 5).astype(int)

# Rush Hour Indicator (Outcome 1: Timing)
df["is_rush"] = df["hour"].isin(RUSH_HOURS).astype(int)

# Time Bins (for Context-Dependent Hypothesis Families)
def get_time_bin(h):
```

```
    for name, lo, hi in TIME_BINS:
        if lo <= h < hi:
            return name
    return "unknown"

df["time_bin"] = df["hour"].apply(get_time_bin)

# Tipping Outcomes (Outcome 2: Tipping Behavior)
# We use tip_rate to control for fare inflation
df["payment_type"] = df["payment_type"].astype(str).str.strip().replace({"Prcard": "Pcard"
df["is_cash"] = (df["payment_type"] == "Cash").astype(int)
df["tip_positive"] = (df["tips"] > 0).astype(int)
df["tip_rate"] = np.where(df["fare"] > 0, df["tips"] / df["fare"], 0.0)

# This prepares the data for the "Hypothesis Family" section of the report

# Filter to Top K Neighborhoods (Spatial Stability)
top_communities = df["pickup_community_area"].value_counts().head(TOP_K).index
df = df.loc[df["pickup_community_area"].isin(top_communities)].copy()

# Define Family 1: Neighborhood Only (Marginal)
# Unit ID = Neighborhood Code
df["unit_f1"] = df["pickup_community_area"].astype(int).astype(str)

# Define Family 2: Neighborhood x Time Context
# Unit ID = Neighborhood_TimeBin (e.g., "8_morning_rush")
df["unit_f2"] = df["unit_f1"] + "_" + df["time_bin"]

print("\n--- Hypothesis Family Prep ---")
print(f"Top {TOP_K} Neighborhoods selected.")
print(f"Family 1 (Neighborhoods) unique units: {df['unit_f1'].nunique()}")
print(f"Family 2 (Nbhd x Time) unique units:   {df['unit_f2'].nunique()}")

# Compare 2014 vs 2019 sample sizes in the final set
print("\nFinal Year Distribution:")
print(df["year"].value_counts())

# Preview Data for Report
display_cols = ["ts_local", "pickup_community_area", "unit_f2", "mph", "tip_rate", "is_rush
print("\nSample Rows:")
display(df[display_cols].head())

# Select top-K neighborhoods by total ride count (both years)
top_counts = df["pickup_community_area"].value_counts()
TOP_NEIGHBORHOODS = top_counts.head(TOP_K).index.astype(int).tolist()

# Filter to top neighborhoods for all subsequent analysis
df_top = df.loc[df["pickup_community_area"].isin(TOP_NEIGHBORHOODS)].copy()

# Family 1: Neighborhood only
df_top["unit_f1"] = df_top["pickup_community_area"].astype(int)

# Family 2: Neighborhood    Time-bin
df_top["unit_f2"] = (
```

```
    df_top["pickup_community_area"].astype(int).astype(str) + "_" + df_top["time_bin"]
)

print(f"\n--- Scientific Units ---")
print(f"Top {TOP_K} neighborhoods selected")
print(f"Family 1 (Neighborhood): {df_top['unit_f1'].nunique()} unique units")
print(f"Family 2 (Nbhd    Time): {df_top['unit_f2'].nunique()} unique units")
print(f"Analysis sample size:    {len(df_top):,} rides")

# Keep a reference to the full cleaned set and the top-neighborhood set
df_all = df.copy()
df = df_top.copy()  # From here on, df = top neighborhoods only
```

## Section 3 Code

```
# 3.1 Ride Volume by Neighborhood & Year
vol = df.groupby(["pickup_community_area", "year"]).size().unstack(fill_value=0)
vol.columns = ["rides_2014", "rides_2019"]
vol["pct_change"] = (vol["rides_2019"] - vol["rides_2014"]) / vol["rides_2014"] * 100
vol = vol.sort_values("rides_2014", ascending=False)
print(vol.to_string())

fig, axes = plt.subplots(1, 2, figsize=(14, 5))
vol_plot = vol[["rides_2014", "rides_2019"]].sort_values("rides_2014", ascending=True)
vol_plot.plot(
    kind="barh",
    ax=axes[0],
    width=0.7,
    color=["#1f77b4", "#4c9ed9"]
)
axes[0].set_title("Ride Volume by Neighborhood & Year")
axes[0].set_xlabel("Number of Rides")
axes[0].set_ylabel("Community Area")

# Percent change: blue scale, red only for largest magnitude change
vol_sorted = vol.sort_values("pct_change")
max_abs = vol_sorted["pct_change"].abs().max()

colors = []
for x in vol_sorted["pct_change"]:
  colors.append("#2c6db2" if x >= 0 else "#6baed6")

axes[1].barh(
    range(len(vol_sorted)),
    vol_sorted["pct_change"],
    color=colors
)
axes[1].set_yticks(range(len(vol_sorted)))
axes[1].set_yticklabels(vol_sorted.index)
axes[1].set_title("% Change in Rides (2014     2019)")
axes[1].set_xlabel("Percent Change")
axes[1].axvline(0, color="#1f3d5a", linewidth=0.8)
```

```
# 3.2 Rush-Hour Share by Neighborhood & Year

fig, ax = plt.subplots(figsize=(8, 5))
rush_sorted = rush_share.sort_values("diff")
colors = ["#9ecae1" if x < 0 else "#08519c" for x in rush_sorted["diff"]]
ax.barh(range(len(rush_sorted)), rush_sorted["diff"], color=colors)
ax.set_yticks(range(len(rush_sorted)))
ax.set_yticklabels(rush_sorted.index)
ax.set_title("Change in Rush-Hour Share (2019     2014)")
ax.set_xlabel("    Rush-Hour Proportion")
ax.axvline(0, color="#08306b", linewidth=0.8)
plt.tight_layout()
plt.show()

# 3.3 Hourly Distribution: 2014 vs 2019
hourly = df.groupby(["year", "hour"]).size().unstack(level=0, fill_value=0)
# Normalize to proportions within each year
hourly_prop = hourly.div(hourly.sum(axis=0), axis=1)

fig, ax = plt.subplots(figsize=(10, 4))
ax.plot(hourly_prop.index, hourly_prop[2014], "o-", label="2014", markersize=4)
ax.plot(hourly_prop.index, hourly_prop[2019], "s-", label="2019", markersize=4)
# Shade rush hours
for h in RUSH_HOURS:
    ax.axvspan(h - 0.5, h + 0.5, alpha=0.1, color="orange")
ax.set_xlabel("Hour of Day (Chicago Local)")
ax.set_ylabel("Proportion of Daily Rides")
ax.set_title("Hourly Ride Distribution (Top 20 Neighborhoods)")
ax.legend()
ax.set_xticks(range(24))
plt.tight_layout()
plt.show()

# 3.4 Tipping behavior

# Among non-cash payments only (cash tips are unrecorded)
df_nc = df.loc[df["payment_type"] != "Cash"].copy()

tip_stats = (
    df_nc.groupby(["pickup_community_area", "year"])
    .agg(
        tip_propensity=("tip_positive", "mean"),
        median_tip_rate=("tip_rate", "median"),
        n_rides=("tip_rate", "size"),
    )
    .reset_index()
)

fig, axes = plt.subplots(1, 2, figsize=(12, 5))

# Tip propensity
tp_wide = tip_stats.pivot(index="pickup_community_area", columns="year", values="tip_prope
tp_wide.columns = ["2014", "2019"]
axes[0].scatter(tp_wide["2014"], tp_wide["2019"], alpha=0.7)
```

```
axes[0].plot([0, 1], [0, 1], "k--", linewidth=0.5)
axes[0].set_xlabel("Tip Propensity 2014")
axes[0].set_ylabel("Tip Propensity 2019")
axes[0].set_title("Tip Propensity (Non-Cash): 2014 vs 2019")
for idx, row in tp_wide.iterrows():
    axes[0].annotate(str(int(idx)), (row["2014"], row["2019"]), fontsize=6)

# Median tip rate
tr_wide = tip_stats.pivot(index="pickup_community_area", columns="year", values="median_ti
tr_wide.columns = ["2014", "2019"]
axes[1].scatter(tr_wide["2014"], tr_wide["2019"], alpha=0.7)
lims = [0, max(tr_wide.max().max() * 1.1, 0.3)]
axes[1].plot(lims, lims, "k--", linewidth=0.5)
axes[1].set_xlabel("Median Tip Rate 2014")
axes[1].set_ylabel("Median Tip Rate 2019")
axes[1].set_title("Median Tip Rate (Non-Cash): 2014 vs 2019")
for idx, row in tr_wide.iterrows():
    axes[1].annotate(str(int(idx)), (row["2014"], row["2019"]), fontsize=6)

plt.tight_layout()
plt.show()

# Cash usage overview
cash_share = (
    df.groupby(["pickup_community_area", "year"])["is_cash"]
    .mean()
    .unstack()
)
cash_share.columns = ["cash_2014", "cash_2019"]
cash_share["diff"] = cash_share["cash_2019"] - cash_share["cash_2014"]

fig, ax = plt.subplots(figsize=(8, 5))
cash_sorted = cash_share.sort_values("diff")
colors = ["#d62728" if x < 0 else "#2ca02c" for x in cash_sorted["diff"]]
ax.barh(range(len(cash_sorted)), cash_sorted["diff"], color=colors)
ax.set_yticks(range(len(cash_sorted)))
ax.set_yticklabels(cash_sorted.index)
ax.set_title("Change in Cash Payment Share (2019     2014)")
ax.set_xlabel("    Cash Proportion")
ax.axvline(0, color="black", linewidth=0.5)
plt.tight_layout()
plt.show()

# Time-bin     neighborhood heatmap

rush_by_timebin = (
    df.groupby(["pickup_community_area", "year", "time_bin"])
    .size()
    .reset_index(name="count")
)
# Compute proportions within each (neighborhood, year)
totals = rush_by_timebin.groupby(["pickup_community_area", "year"])["count"].transform("sum
rush_by_timebin["prop"] = rush_by_timebin["count"] / totals
```

```
# Pivot for 2014 and 2019, compute difference
for yr in [2014, 2019]:
    piv = rush_by_timebin.loc[rush_by_timebin["year"] == yr].pivot(
        index="pickup_community_area", columns="time_bin", values="prop"
    )
    if yr == 2014:
        piv_14 = piv
    else:
        piv_19 = piv

diff_heatmap = (piv_19 - piv_14).reindex(
    columns=["late_night", "morning_rush", "midday", "evening_rush", "night"]
)

fig, ax = plt.subplots(figsize=(10, 7))
sns.heatmap(
    diff_heatmap, annot=True, fmt=".3f", cmap="RdBu_r", center=0,
    linewidths=0.5, ax=ax
)
ax.set_title("Change in Time-Bin Share (2019      2014) by Neighborhood")
ax.set_ylabel("Community Area")
ax.set_xlabel("Time Bin")
plt.tight_layout()
plt.show()
```

### Section 4.1 Code

```
library(tidyverse)
library(ggplot2)
library(dplyr)
library(haven)
library(glmnet)
library(lmtest)
library(car)
library(rdrobust)
library(rddensity)
library(sandwich)
library(lubridate)

taxi_dat <- read_csv('taxidata-1.csv')

taxi <- taxi_dat %>%
  mutate(
    trip_start_timestamp = ymd_hms(trip_start_timestamp, quiet = TRUE),
    year = year(trip_start_timestamp),
    hour = hour(trip_start_timestamp)
  ) %>%
  # Keep only 2014 + 2019
  filter(year %in% c(2014, 2019)) %>%
  # Keep only valid neighborhood id
  mutate(pickup_community_area = as.integer(pickup_community_area)) %>%
  filter(!is.na(pickup_community_area), pickup_community_area > 0) %>%
```

```
  mutate(across(
    c(trip_seconds, trip_miles, fare, extras, trip_total, tips),
    ~ suppressWarnings(as.numeric(.x))
  )) %>%
  # No NA
  filter(
    !is.na(trip_seconds), !is.na(trip_miles), !is.na(trip_total),
    trip_seconds > 0, trip_miles > 0, trip_total > 0
  ) %>%
  # Clean for outliers
  filter(
    trip_miles <= 100,
    trip_seconds <= 4 * 3600,
    trip_total <= 500
  ) %>%
  # Rush hour
  mutate(
    rush = as.integer((hour >= 7 & hour <= 9) | (hour >= 16 & hour <= 18))
  )

agg_table <- taxi %>%
  group_by(pickup_community_area, year) %>%
  summarise(
    num_rides = n(),
    num_rush_rides = sum(rush, na.rm = TRUE),
    p = num_rush_rides / num_rides,
    .groups = "drop"
  ) %>%
  pivot_wider(
    names_from = year,
    values_from = c(num_rides, num_rush_rides, p),
    names_sep = "_"
  ) %>%
  filter(!is.na(num_rides_2014), !is.na(num_rides_2019)) %>%
  mutate(delta = p_2019 - p_2014)

get_pvals <- function(s1, n1, s2, n2) {
  p_pool <- (s1 + s2) / (n1 + n2)
  se <- sqrt(p_pool * (1 - p_pool) * (1 / n1 + 1 / n2))
  z <- (s2 / n2 - s1 / n1) / se
  2 * pnorm(-abs(z))
}

n_grid <- c(0, 10, 25, 50, 100, 200, 400)
q_level <- 0.10

get_table_pvals <- function(ny_tbl, n_grid, q_level = 0.10) {
  map_dfr(n_grid, function(nmin) {
    dat <- ny_tbl %>%
      filter(num_rides_2014 >= nmin, num_rides_2019 >= nmin) %>%
      mutate(
        pval = get_pvals(num_rush_rides_2014, num_rides_2014,
                         num_rush_rides_2019, num_rides_2019),
        qval = p.adjust(pval, method = "BH"),
```

```
        discovered = (qval <= q_level)
      ) %>%
      mutate(n_min = nmin)
    dat
  })
}
result_nmin <- get_table_pvals(agg_table, n_grid, q_level)
summary_nmin <- result_nmin %>%
  group_by(n_min) %>%
  summarise(
    m_tested = n(),
    n_disc = sum(discovered, na.rm = TRUE),
    .groups = "drop"
  )

result_nmin %>%
  filter(n_min == 0) %>%
  summarise(
    n_rows = n(),
    n_pval_na = sum(is.na(pval)),
    n_qval_na = sum(is.na(qval)),
    n_disc_na = sum(is.na(discovered))
  )


df2 <- result_nmin %>%
  filter(n_min == 100) %>%
  select(pickup_community_area, num_rides_2014, num_rides_2019,
         p_2014, p_2019, delta, pval, qval, discovered)

K <- 20  # how many neighborhoods to show

topK <- df2 %>%
  arrange(pval) %>%
  slice_head(n = K) %>%
  mutate(
    label = paste0("Neighborhood ", pickup_community_area),
    label = forcats::fct_reorder(label, delta),
    sig_BH = if_else(discovered, "BH significant", "Not significant")
  )

ggplot(topK, aes(x = label, y = delta, fill = sig_BH)) +
  geom_col() +
  geom_hline(yintercept = 0) +
  coord_flip(clip = "off") +
  labs(
    x = "Pickup community area",
    y = "Change in rush-hour share (p2019     p2014)",
    title = paste0("Top ", K, " neighborhoods by smallest p-values (n_min = 100)"),
    fill = ""
  ) +
  theme(plot.margin = margin(5.5, 80, 5.5, 5.5))

# BY
```

```r
# BY-adjusted p-values to the n_min results
result_nmin_BY <- result_nmin %>%
  mutate(
    qval_BY = p.adjust(pval, method = "BY"),
    discovered_BY = (qval_BY <= q_level)
  )

# BH vs BY discoveries for each n_min
summary_nmin_BH_BY <- result_nmin_BY %>%
  group_by(n_min) %>%
  summarise(
    m_tested = n(),
    n_disc_BH = sum(discovered, na.rm = TRUE),
    n_disc_BY = sum(discovered_BY, na.rm = TRUE),
    .groups = "drop"
  )

df2_BY <- result_nmin_BY %>%
  filter(n_min == 100) %>%
  filter(is.finite(pval)) %>%
  select(
    pickup_community_area,
    num_rides_2014, num_rides_2019,
    p_2014, p_2019, delta,
    pval,
    qval_BH = qval, discovered_BH = discovered,
    qval_BY, discovered_BY
  )


# plot w/ colors showing BH, BY, both etc
topK_BY <- df2_BY %>%
  arrange(pval) %>%
  slice_head(n = K) %>%
  mutate(
    label = paste0("Neighborhood ", pickup_community_area),
    label = forcats::fct_reorder(label, delta),
    sig = case_when(
      discovered_BY ~ "BY",
      discovered_BH ~ "BH only",
      TRUE ~ "none"
    )
  )

ggplot(topK_BY, aes(x = label, y = delta, fill = sig)) +
  geom_col() +
  geom_hline(yintercept = 0) +
  coord_flip(clip = "off") +
  labs(
    x = "Pickup community area",
    y = "Change in rush-hour share (p2019 - p2014)",
    title = paste0("Top ", K, " neighborhoods by smallest p-values (n_min = 100)"),
    fill = "Significant under"
```

```
  ) +
  theme(plot.margin = margin(5.5, 80, 5.5, 5.5))
```

## Section 4.2 Code

```
# 5 time bins
taxi2 <- taxi %>%
  mutate(
    time_bin = case_when(
      hour >= 0  & hour <= 5  ~ "late_night",      # 00:00  05  :59
      hour >= 6  & hour <= 9  ~ "morning_rush",    # 06:00  09  :59
      hour >= 10 & hour <= 15 ~ "midday",          # 10:00  15  :59
      hour >= 16 & hour <= 18 ~ "evening_rush",    # 16:00  18  :59
      hour >= 19 & hour <= 23 ~ "night",           # 19:00  23  :59
      TRUE ~ NA_character_
    ),
    time_bin = factor(time_bin, levels = c("late_night","morning_rush","midday","evening_ru
  ) %>%
  filter(!is.na(time_bin))

totals_by_yr <- taxi2 %>%
  group_by(pickup_community_area, year) %>%
  summarise(n_total = n(), .groups = "drop")

# By time of day, by year, by neighborhood
bin_counts <- taxi2 %>%
  group_by(pickup_community_area, year, time_bin) %>%
  summarise(n_bin = n(), .groups = "drop")

# Combining two above tables
comb_table <- bin_counts %>%
  left_join(totals_by_yr, by = c("pickup_community_area","year")) %>%
  pivot_wider(
    names_from = year,
    values_from = c(n_bin, n_total),
    names_sep = "_",
    values_fill = 0
  ) %>%
  filter(n_total_2014 > 0, n_total_2019 > 0) %>%
  mutate(
    p_2014 = n_bin_2014 / n_total_2014,
    p_2019 = n_bin_2019 / n_total_2019,
    delta  = p_2019 - p_2014
  )

# pooled z-test
get_pvals <- function(s1, n1, s2, n2) {
  p_pool <- (s1 + s2) / (n1 + n2)
  se <- sqrt(p_pool * (1 - p_pool) * (1 / n1 + 1 / n2))
  z <- (s2 / n2 - s1 / n1) / se
  2 * pnorm(-abs(z))
}
```

```r
n_vals <- c(0, 10, 25, 50, 100, 200, 400)
q_level <- 0.10

get_table_pvals_2 <- function(comb_table, n_grid, q_level = 0.10) {
  purrr::map_dfr(n_grid, function(nmin) {
    dat <- comb_table %>%
      filter(n_bin_2014 >= nmin, n_bin_2019 >= nmin) %>%
      mutate(
        pval = get_pvals(n_bin_2014, n_total_2014, n_bin_2019, n_total_2019),
        q_BH = p.adjust(pval, method = "BH"),
        q_BY = p.adjust(pval, method = "BY"),
        disc_BH = (q_BH <= q_level),
        disc_BY = (q_BY <= q_level),
        n_min = nmin
      )
    dat
  })
}

result_2 <- get_table_pvals_2(comb_table, n_vals, q_level)

# Summary table for output
summary_2 <- result_2 %>%
  group_by(n_min) %>%
  summarise(
    m_tested = n(),
    n_disc_BH = sum(disc_BH, na.rm = TRUE),
    n_disc_BY = sum(disc_BY, na.rm = TRUE),
    .groups = "drop"
  )
```

## Section 5.1 Code

```r
## SECTION 5: ADDITIONAL OUTCOMES
NEIGHBORHOODS_ALL <- sort(unique(df$pickup_community_area))
NEIGHBORHOODS_ALL <- NEIGHBORHOODS_ALL[!is.na(NEIGHBORHOODS_ALL)]

RNG_SEED <- 27850
set.seed(RNG_SEED)

# 5.1 Cash payment share (F1)
res_cash_f1 <- f1_ztest_binary(df, NEIGHBORHOODS_ALL, "is_cash", nmin = NMIN)
res_cash_f1 <- add_corrections(res_cash_f1)
cat("\n[Cash F1] ", nrow(res_cash_f1), " hypotheses\n", sep="")
discovery_summary(res_cash_f1, "Cash F1 (prop.test)")

# (A) Cash effect barplot
if (nrow(res_cash_f1) > 0) {
  cash_sorted <- res_cash_f1[order(res_cash_f1$effect), ]
  cols <- ifelse(cash_sorted$effect < 0, "red3", "darkgreen")

  H <- max(900, 18 * nrow(cash_sorted))  # scale height for 77 labels
```

```
  png("fig_5_1_cash_effect_bar.png", width = 1500, height = H, res = 150)
  par(mar = c(6, 9, 4, 2), xpd = NA)
  barplot(cash_sorted$effect, horiz = TRUE, las = 1, col = cols,
          names.arg = cash_sorted$pickup_community_area,
          main = "Cash payment share: effect by neighborhood (2019 - 2014)",
          xlab = "Effect (p2019 - p2014)",
          cex.names = 0.8)
  abline(v = 0, lwd = 2)
  dev.off()
}

# (B) Cash p-value histogram
if (nrow(res_cash_f1) > 0) {
  png("fig_5_1_cash_pval_hist.png", width = 1100, height = 650, res = 150)
  par(mar = c(5, 5, 4, 2), xpd = NA)
  hist(res_cash_f1$pval, breaks = 20, col = "gray85", border = "white",
       main = "Cash F1: p-value histogram", xlab = "p-value")
  abline(h = length(res_cash_f1$pval)/20, col = "red3", lty = 2, lwd = 2)
  dev.off()
}
```

## Section 5.2 Code

```
    # 5.2 Tip propensity (F2)

df_noncash <- df[df$payment_type != "Cash" & !is.na(df$payment_type), ]

res_tip_f2 <- f2_ztest_binary(df_noncash, NEIGHBORHOODS_ALL, "tip_positive", nmin = NMIN)
res_tip_f2 <- add_corrections(res_tip_f2)
cat("\n[Tip propensity F2] ", nrow(res_tip_f2), " hypotheses\n", sep="")
discovery_summary(res_tip_f2, "Tip F2 (prop.test)")

# (A) Tip effect heatmap
if (nrow(res_tip_f2) > 0) {
  bins_order <- c("late_night", "morning_rush", "midday", "evening_rush", "night")

  eff_mat <- with(res_tip_f2, tapply(effect, list(pickup_community_area, time_bin), mean, n
  eff_mat[is.na(eff_mat)] <- 0

  # keep consistent column order when available
  bins_keep <- bins_order[bins_order %in% colnames(eff_mat)]
  eff_mat <- eff_mat[, bins_keep, drop = FALSE]

  # order neighborhoods by max abs effect
  ord <- order(apply(abs(eff_mat), 1, max), decreasing = TRUE)
  eff_mat <- eff_mat[ord, , drop = FALSE]

  zmax <- max(abs(eff_mat))
  if (!is.finite(zmax) || zmax == 0) zmax <- 1e-6
  cols <- grDevices::colorRampPalette(c("red3", "white", "darkgreen"))(101)

  H <- max(900, 16 * nrow(eff_mat))
```

```
  png("fig_5_2_tip_effect_heatmap.png", width = 1400, height = H, res = 150)
  par(mar = c(7, 9, 4, 2), xpd = NA)

  image(1:ncol(eff_mat), 1:nrow(eff_mat),
        t(eff_mat[nrow(eff_mat):1, , drop = FALSE]),
        col = cols, zlim = c(-zmax, zmax),
        axes = FALSE,
        main = "Tip propensity (non-cash, F2): effect heatmap (2019 - 2014)")

  axis(1, at = 1:ncol(eff_mat), labels = colnames(eff_mat), las = 2)
  axis(2, at = 1:nrow(eff_mat), labels = rev(rownames(eff_mat)), las = 1, cex.axis = 0.65)
  dev.off()
}

# (B) Tip p-value histogram
if (nrow(res_tip_f2) > 0) {
  png("fig_5_2_tip_pval_hist.png", width = 1100, height = 650, res = 150)
  par(mar = c(5, 5, 4, 2), xpd = NA)
  hist(res_tip_f2$pval, breaks = 20, col = "gray85", border = "white",
       main = "Tip propensity F2: p-value histogram", xlab = "p-value")
  abline(h = length(res_tip_f2$pval)/20, col = "red3", lty = 2, lwd = 2)
  dev.off()
}
```

## Section 5.3 Code

```
# 5.3 Joint correction
joint <- rbind(
  data.frame(outcome = "rush_f1",
             pickup_community_area = res_rush_f1_z$pickup_community_area,
             context = NA,
             pval = res_rush_f1_z$pval),
  data.frame(outcome = "cash_f1",
             pickup_community_area = res_cash_f1$pickup_community_area,
             context = NA,
             pval = res_cash_f1$pval),
  data.frame(outcome = "tip_f2",
             pickup_community_area = res_tip_f2$pickup_community_area,
             context = res_tip_f2$time_bin,
             pval = res_tip_f2$pval)
)

joint <- add_corrections(joint, pcol = "pval")
cat("\n[Joint family] m=", nrow(joint), " across outcomes\n", sep="")
discovery_summary(joint, "Joint (all outcomes)")

# Joint discoveries barplot
disc <- c(
  Bonferroni = sum(joint$rej_bonf, na.rm = TRUE),
  Holm       = sum(joint$rej_holm, na.rm = TRUE),
  BH         = sum(joint$rej_bh, na.rm = TRUE),
  BY         = sum(joint$rej_by, na.rm = TRUE),
  Storey     = sum(joint$rej_storey, na.rm = TRUE)
```

```
)

png("fig_5_3_joint_discoveries.png", width = 950, height = 650, res = 150)
par(mar = c(5, 5, 4, 2), xpd = NA)
barplot(disc, las = 1, col = "gray75",
        main = "Joint family: discoveries by correction (alpha = 0.10)",
        ylab = "Number of discoveries")
dev.off()
```

## Section 6.1 Code

```
def add_corrections(res, pcol="pval", alpha=ALPHA):
    """Add Bonferroni, Holm, BH, BY, and Storey q-values."""
    out = res.copy()
    p = out[pcol].to_numpy().astype(float)
    m = len(p)
    if m == 0:
        return out

    out["p_bonf"] = np.minimum(p * m, 1.0)
    out["rej_bonf"] = out["p_bonf"] <= alpha

    out["rej_holm"], out["p_holm"], _, _ = multipletests(p, alpha=alpha, method="holm")
    out["rej_bh"], out["q_bh"], _, _ = multipletests(p, alpha=alpha, method="fdr_bh")
    out["rej_by"], out["q_by"], _, _ = multipletests(p, alpha=alpha, method="fdr_by")

    lam = 0.5
    pi0 = np.clip(np.mean(p > lam) / (1 - lam), 0.0, 1.0)
    order = np.argsort(p)
    ps = p[order]
    q = pi0 * m * ps / np.arange(1, m + 1)
    q = np.minimum.accumulate(q[::-1])[::-1]
    q = np.clip(q, 0, 1)
    qvals = np.empty_like(q)
    qvals[order] = q

    out["q_storey"] = qvals
    out["rej_storey"] = out["q_storey"] <= alpha
    out["pi0_storey"] = pi0
    return out


def f1_ztest_binary(df_in, neighborhoods, ycol, year_col="year",
                    year_vals=(2014, 2019), nmin=NMIN):
    """
    Two-proportion z-test per neighborhood.
    Generalized to accept arbitrary year_col and year_vals for placebo use.
    """
    y_old, y_new = year_vals
    rows = []
    for a in neighborhoods:
        da = df_in.loc[df_in["pickup_community_area"] == a]
```

```
        d_old = da.loc[da[year_col] == y_old, ycol]
        d_new = da.loc[da[year_col] == y_new, ycol]
        n_old, n_new = len(d_old), len(d_new)
        if min(n_old, n_new) < nmin:
            continue

        k_old, k_new = int(d_old.sum()), int(d_new.sum())
        stat, p = proportions_ztest([k_new, k_old], [n_new, n_old],
                                    alternative="two-sided")
        rows.append({
            "pickup_community_area": a,
            "n_old": n_old, "n_new": n_new,
            "p_old": k_old / n_old, "p_new": k_new / n_new,
            "effect": k_new / n_new - k_old / n_old,
            "zstat": float(stat),
            "pval": float(p),
        })
    return pd.DataFrame(rows)


def pval_diagnostics(pvals, title, save_prefix=None):
    """P-value histogram + QQ plot."""
    p = np.asarray(pvals, float)
    p = p[np.isfinite(p)]
    if len(p) == 0:
        print(f"No finite p-values for: {title}")
        return

    fig, axes = plt.subplots(1, 2, figsize=(10, 4))

    axes[0].hist(p, bins=20, edgecolor="white", alpha=0.8)
    axes[0].axhline(len(p) / 20, color="red", linestyle="--",
                    linewidth=1, label="Uniform expectation")
    axes[0].set_title(f"{title}  P-value Histogram")
    axes[0].set_xlabel("p-value")
    axes[0].set_ylabel("Count")
    axes[0].legend(fontsize=8)

    p_sorted = np.sort(p)
    u = (np.arange(1, len(p_sorted) + 1) - 0.5) / len(p_sorted)
    axes[1].scatter(u, p_sorted, s=15, alpha=0.7)
    axes[1].plot([0, 1], [0, 1], "r--", linewidth=1)
    axes[1].set_title(f"{title}  QQ vs Uniform")
    axes[1].set_xlabel("Uniform quantiles")
    axes[1].set_ylabel("Observed p quantiles")

    plt.tight_layout()
    if save_prefix:
        plt.savefig(f"{save_prefix}.png", bbox_inches="tight")
    plt.show()


def discovery_summary(res, label=""):
    """Print compact discovery counts."""
```

```
    methods = ["rej_bonf", "rej_holm", "rej_bh", "rej_by", "rej_storey"]
    counts = {m.replace("rej_", "").upper(): int(res[m].sum())
              for m in methods if m in res.columns}
    print(f"  {label} | m={len(res)} | Discoveries: {counts}")
    return counts


# placebo A: WITHIN-YEAR RANDOM SPLIT
N_PLACEBO_REPS = 50  # number of random splits per year

def run_within_year_placebo(df_full, source_year, ycol, n_reps=N_PLACEBO_REPS,
                            nmin=NMIN, seed=RNG_SEED):
    """
    Split one year's data into two random halves, run F1 z-test pipeline.
    Repeat n_reps times. Returns DataFrame of per-rep discovery counts.
    """
    df_year = df_full.loc[df_full["year"] == source_year].copy()
    rng_local = np.random.default_rng(seed)

    rep_results = []
    all_pvals = []  # collect all p-values across reps for diagnostics

    for r in range(n_reps):
        # Random binary assignment within each neighborhood
        # (preserves neighborhood sample sizes in each arm)
        df_year["placebo_arm"] = rng_local.integers(0, 2, size=len(df_year))

        res = f1_ztest_binary(
            df_year, TOP_NEIGHBORHOODS, ycol,
            year_col="placebo_arm", year_vals=(0, 1),
            nmin=nmin // 2  # each arm has ~half the rides, so halve nmin
        )
        if len(res) == 0:
            rep_results.append({
                "rep": r, "year": source_year,
                "m": 0, "disc_bh": 0, "disc_bonf": 0
            })
            continue

        res = add_corrections(res)
        all_pvals.extend(res["pval"].tolist())

        rep_results.append({
            "rep": r,
            "year": source_year,
            "m": len(res),
            "disc_bh": int(res["rej_bh"].sum()),
            "disc_bonf": int(res["rej_bonf"].sum()),
            "disc_storey": int(res["rej_storey"].sum()),
        })

    return pd.DataFrame(rep_results), np.array(all_pvals)
```

```
# Run placebo on each year
for yr in [2014, 2019]:
    placebo_df, placebo_pvals = run_within_year_placebo(
        df, source_year=yr, ycol="is_rush",
        n_reps=N_PLACEBO_REPS, seed=RNG_SEED + yr
    )

    print(f"\n Within-{yr} Split ")
    print(f"  Mean hypotheses per rep:    {placebo_df['m'].mean():.1f}")
    print(f"  Mean BH discoveries:        {placebo_df['disc_bh'].mean():.2f}")
    print(f"  Mean Bonferroni discoveries: {placebo_df['disc_bonf'].mean():.2f}")
    print(f"  Reps with  1  BH discovery:  {(placebo_df['disc_bh'] > 0).sum()} / {N_PLACE
    print(f"  Expected false disc (BH):   ~{ALPHA * placebo_df['m'].mean():.1f} "
          f"(  m = {ALPHA}  {placebo_df['m'].mean():.0f})")

    # Plot distribution of false discovery counts
    fig, axes = plt.subplots(1, 2, figsize=(12, 4))

    axes[0].hist(placebo_df["disc_bh"], bins=range(0, placebo_df["disc_bh"].max() + 3),
                 edgecolor="white", alpha=0.8, align="left")
    axes[0].axvline(placebo_df["disc_bh"].mean(), color="red", linestyle="--",
                    label=f"Mean = {placebo_df['disc_bh'].mean():.2f}")
    axes[0].set_title(f"Placebo A ({yr}): BH False Discoveries per Rep")
    axes[0].set_xlabel("# BH Discoveries (all false positives)")
    axes[0].set_ylabel("Count of reps")
    axes[0].legend(fontsize=8)

    # P-value diagnostics across all reps (pooled)
    if len(placebo_pvals) > 0:
        axes[1].hist(placebo_pvals, bins=20, edgecolor="white", alpha=0.8, density=True)
        axes[1].axhline(1.0, color="red", linestyle="--", label="Uniform density")
        axes[1].set_title(f"Placebo A ({yr}): Pooled P-value Distribution")
        axes[1].set_xlabel("p-value")
        axes[1].set_ylabel("Density")
        axes[1].legend(fontsize=8)
    plt.tight_layout()
    plt.savefig(f"fig_6_1a_placebo_within_{yr}.png", bbox_inches="tight")
    plt.show()

#  placebo B: YEAR-LABEL PERMUTATION NULL

N_PERM_REPS = 50

def run_permutation_null(df_full, ycol, n_reps=N_PERM_REPS, nmin=NMIN, seed=RNG_SEED):
    rng_local = np.random.default_rng(seed)
    df_work = df_full.copy()
    rep_results = []
    all_pvals = []

    for r in range(n_reps):
        # Shuffle year labels globally
        df_work["year_perm"] = rng_local.permutation(df_work["year"].to_numpy())

        res = f1_ztest_binary(
```

```
            df_work, TOP_NEIGHBORHOODS, ycol,
            year_col="year_perm", year_vals=(2014, 2019),
            nmin=nmin
        )
        if len(res) == 0:
            rep_results.append({"rep": r, "m": 0, "disc_bh": 0, "disc_bonf": 0})
            continue

        res = add_corrections(res)
        all_pvals.extend(res["pval"].tolist())

        rep_results.append({
            "rep": r,
            "m": len(res),
            "disc_bh": int(res["rej_bh"].sum()),
            "disc_bonf": int(res["rej_bonf"].sum()),
            "disc_storey": int(res["rej_storey"].sum()),
        })
    return pd.DataFrame(rep_results), np.array(all_pvals)

perm_null_df, perm_null_pvals = run_permutation_null(
    df, ycol="is_rush", n_reps=N_PERM_REPS, seed=RNG_SEED + 999
)

# Get real discovery count for comparison
res_real = f1_ztest_binary(df, TOP_NEIGHBORHOODS, "is_rush", nmin=NMIN)
res_real = add_corrections(res_real)
real_disc_bh = int(res_real["rej_bh"].sum())

print(f"\n Placebo B: Permutation Null ")
print(f"  Mean hypotheses per rep:     {perm_null_df['m'].mean():.1f}")
print(f"  Mean BH discoveries (null):  {perm_null_df['disc_bh'].mean():.2f}")
print(f"  Max BH discoveries (null):   {perm_null_df['disc_bh'].max()}")
print(f"  Real BH discoveries:         {real_disc_bh}")
print(f"  Permutation p-value:         "
      f"{(1 + (perm_null_df['disc_bh'] >= real_disc_bh).sum()) / (N_PERM_REPS + 1):.4f}")
print(f"  (fraction of null reps with   {real_disc_bh} discoveries)")

# Plot
fig, axes = plt.subplots(1, 2, figsize=(12, 4))

axes[0].hist(perm_null_df["disc_bh"], bins=range(0, max(perm_null_df["disc_bh"].max(), rea
             edgecolor="white", alpha=0.8, align="left", label="Permutation null")
axes[0].axvline(real_disc_bh, color="red", linewidth=2, linestyle="--",
                label=f"Real data: {real_disc_bh}")
axes[0].set_title("Placebo B: Null Discovery Distribution vs Real")
axes[0].set_xlabel("# BH Discoveries")
axes[0].set_ylabel("Count of permutation reps")
axes[0].legend(fontsize=8)

if len(perm_null_pvals) > 0:
    axes[1].hist(perm_null_pvals, bins=20, edgecolor="white", alpha=0.8, density=True)
    axes[1].axhline(1.0, color="red", linestyle="--", label="Uniform density")
    axes[1].set_title("Placebo B: Pooled Null P-values")
```

```
    axes[1].set_xlabel("p-value")
    axes[1].set_ylabel("Density")
    axes[1].legend(fontsize=8)

plt.tight_layout()
plt.savefig("fig_6_1b_placebo_perm_null.png", bbox_inches="tight")
plt.show()

# PLACEBOS FOR CASH OUTCOME
for yr in [2014, 2019]:
    placebo_df_cash, _ = run_within_year_placebo(
        df, source_year=yr, ycol="is_cash",
        n_reps=30, seed=RNG_SEED + yr + 5000
    )
    print(f"  Cash within-{yr}: mean BH disc = {placebo_df_cash['disc_bh'].mean():.2f}, "
          f"reps with  1  disc = {(placebo_df_cash['disc_bh'] > 0).sum()}/30")

perm_cash_df, _ = run_permutation_null(
    df, ycol="is_cash", n_reps=30, seed=RNG_SEED + 7777
)
res_cash_real = f1_ztest_binary(df, TOP_NEIGHBORHOODS, "is_cash", nmin=NMIN)
res_cash_real = add_corrections(res_cash_real)
print(f"  Cash perm null: mean BH disc = {perm_cash_df['disc_bh'].mean():.2f}")
print(f"  Cash REAL BH disc = {int(res_cash_real['rej_bh'].sum())}")
```

## Section 6.2 Code

```
STABILITY_REPS = 30
STABILITY_FRAC = 0.5
def get_discovery_set(df_in, ycol, neighborhoods, nmin, alpha=ALPHA, method="bh"):
    res = f1_ztest_binary(df_in, neighborhoods, ycol, nmin=nmin)
    if len(res) == 0:
        return set(), res
    res = add_corrections(res, alpha=alpha)
    col = f"rej_{method}"
    return set(res.loc[res[col], "pickup_community_area"].tolist()), res


def run_stability(df_full, ycol, neighborhoods, nmin=NMIN, n_reps=STABILITY_REPS,
                  frac=STABILITY_FRAC, seed=RNG_SEED):
    full_disc, _ = get_discovery_set(df_full, ycol, neighborhoods, nmin)

    rng_local = np.random.default_rng(seed)
    rep_rows = []
    nbhd_counts = Counter()  # how often each neighborhood is discovered

    for r in range(n_reps):
        d14 = df_full.loc[df_full["year"] == 2014].sample(
            frac=frac, random_state=rng_local.integers(0, 2**31))
        d19 = df_full.loc[df_full["year"] == 2019].sample(
            frac=frac, random_state=rng_local.integers(0, 2**31))
        dsub = pd.concat([d14, d19], ignore_index=True)
```

```
        sub_disc, _ = get_discovery_set(dsub, ycol, neighborhoods, nmin)

        for a in sub_disc:
            nbhd_counts[a] += 1

        overlap = full_disc & sub_disc
        union = full_disc | sub_disc
        jaccard = len(overlap) / len(union) if len(union) > 0 else 1.0

        rep_rows.append({
            "rep": r,
            "n_disc_sub": len(sub_disc),
            "n_overlap": len(overlap),
            "n_full_only": len(full_disc - sub_disc),
            "n_sub_only": len(sub_disc - full_disc),
            "jaccard": jaccard,
        })
    return pd.DataFrame(rep_rows), full_disc, nbhd_counts


# Run for rush-hour (primary outcome)
stab_rush, full_disc_rush, nbhd_freq_rush = run_stability(
    df, "is_rush", TOP_NEIGHBORHOODS, seed=RNG_SEED
)
# Run for cash (secondary outcome)
stab_cash, full_disc_cash, nbhd_freq_cash = run_stability(
    df, "is_cash", TOP_NEIGHBORHOODS, seed=RNG_SEED + 2000
)


for label, stab_df, full_disc, nbhd_freq in [
    ("Rush-Hour F1", stab_rush, full_disc_rush, nbhd_freq_rush),
    ("Cash F1", stab_cash, full_disc_cash, nbhd_freq_cash),
]:
    print(f"\n--- {label} Stability ({STABILITY_REPS} reps, {int(STABILITY_FRAC*100)}% subs
    print(f"  Full-sample discoveries:  {len(full_disc)} {sorted(full_disc)}")
    print(f"  Mean subsample disc.:     {stab_df['n_disc_sub'].mean():.1f}    {stab_df['n_
    print(f"  Mean Jaccard overlap:     {stab_df['jaccard'].mean():.3f}    {stab_df['jaccar
    print(f"  Mean overlap with full:   {stab_df['n_overlap'].mean():.1f}")
    print(f"  Mean full-only (missed):  {stab_df['n_full_only'].mean():.1f}")
    print(f"  Mean sub-only (new):      {stab_df['n_sub_only'].mean():.1f}")
    print(f"  Per-neighborhood recovery rate:")
    for a in sorted(full_disc):
        rate = nbhd_freq.get(a, 0) / STABILITY_REPS
        print(f"    Area {a:>3d}: recovered in {nbhd_freq.get(a, 0):>2d}/{STABILITY_REPS} "
              f"({rate:.0%})")

# viz
def plot_stability(stab_df, full_disc, nbhd_freq, outcome_label, n_reps, save_path):
    fig = plt.figure(figsize=(15, 5))
    gs = gridspec.GridSpec(1, 3, width_ratios=[1.2, 1, 1], wspace=0.3)

    # Per-neighborhood recovery rate
    ax0 = fig.add_subplot(gs[0])
```

```
# Include all neighborhoods that were EVER discovered (full or any subsample)
all_discovered = sorted(set(full_disc) | set(nbhd_freq.keys()))
rates = []
for a in all_discovered:
    rate = nbhd_freq.get(a, 0) / n_reps
    in_full = a in full_disc
    rates.append((a, rate, in_full))

rates.sort(key=lambda x: x[1], reverse=True)
names = [str(r[0]) for r in rates]
values = [r[1] for r in rates]
colors = ["#2166ac" if r[2] else "#b2182b" for r in rates]

bars = ax0.barh(range(len(rates)), values, color=colors, edgecolor="white", height=0.7
ax0.set_yticks(range(len(rates)))
ax0.set_yticklabels(names, fontsize=8)
ax0.set_xlabel("Recovery Rate (fraction of subsamples)")
ax0.set_title("Per-Neighborhood Stability")
ax0.axvline(0.5, color="gray", linestyle=":", linewidth=0.8, alpha=0.7)
ax0.set_xlim(0, 1.05)
ax0.invert_yaxis()

legend_elements = [
    Patch(facecolor="#2166ac", label="In full-sample discoveries"),
    Patch(facecolor="#b2182b", label="Only in subsamples"),
]
ax0.legend(handles=legend_elements, fontsize=7, loc="lower right")

# Discovery count distribution
ax1 = fig.add_subplot(gs[1])

disc_counts = stab_df["n_disc_sub"].values
bins = np.arange(disc_counts.min() - 0.5, max(disc_counts.max(), len(full_disc)) + 1.5
ax1.hist(disc_counts, bins=bins, edgecolor="white", alpha=0.8, color="#4393c3")
ax1.axvline(len(full_disc), color="#d6604d", linewidth=2, linestyle="--",
            label=f"Full sample: {len(full_disc)}")
ax1.axvline(stab_df["n_disc_sub"].mean(), color="#2166ac", linewidth=1.5, linestyle=":
            label=f"Mean subsample: {stab_df['n_disc_sub'].mean():.1f}")
ax1.set_xlabel("# BH Discoveries")
ax1.set_ylabel("Count of subsamples")
ax1.set_title("Discovery Count Distribution")
ax1.legend(fontsize=7)

# Jaccard distribution
ax2 = fig.add_subplot(gs[2])

jaccards = stab_df["jaccard"].values
ax2.hist(jaccards, bins=12, edgecolor="white", alpha=0.8, color="#4393c3")
ax2.axvline(jaccards.mean(), color="#2166ac", linewidth=1.5, linestyle=":",
            label=f"Mean: {jaccards.mean():.3f}")
ax2.set_xlabel("Jaccard Index (overlap with full sample)")
ax2.set_ylabel("Count of subsamples")
ax2.set_title("Jaccard Overlap Distribution")
```

```
    ax2.legend(fontsize=7)
    ax2.set_xlim(0, 1.05)

    fig.suptitle(f"Stability Analysis: {outcome_label} (BH,     = {ALPHA})",
                 fontsize=13, y=1.02)
    plt.tight_layout()
    plt.savefig(save_path, bbox_inches="tight", dpi=150)
    plt.show()


plot_stability(stab_rush, full_disc_rush, nbhd_freq_rush,
               "Rush-Hour F1", STABILITY_REPS)

plot_stability(stab_cash, full_disc_cash, nbhd_freq_cash,
               "Cash F1", STABILITY_REPS)
```

### Section 6.3 Code

```
def comprehensive_pval_panel(results_dict, save_path):
    n_analyses = len(results_dict)
    fig, axes = plt.subplots(n_analyses, 2, figsize=(11, 3.2 * n_analyses))
    if n_analyses == 1:
        axes = axes.reshape(1, -1)

    for i, (label, pvals) in enumerate(results_dict.items()):
        p = np.asarray(pvals, float)
        p = p[np.isfinite(p)]
        m = len(p)

        # histogram
        ax_h = axes[i, 0]
        ax_h.hist(p, bins=20, edgecolor="white", alpha=0.8, color="#4393c3", density=True)
        ax_h.axhline(1.0, color="#d6604d", linestyle="--", linewidth=1, label="Uniform")
        ax_h.set_xlabel("p-value")
        ax_h.set_ylabel("Density")
        ax_h.set_title(f"{label} (m = {m})")
        ax_h.legend(fontsize=7)
        ax_h.set_xlim(-0.02, 1.02)

        # KS test against Uniform(0,1)
        ks_stat, ks_pval = stats.kstest(p, "uniform")
        # Fraction below 0.05 (expected: 5%)
        frac_005 = np.mean(p < 0.05)
        frac_010 = np.mean(p < 0.10)

        # Annotate inside histogram
        summary_text = (
            f"KS stat: {ks_stat:.3f} (p={ks_pval:.3f})\n"
            f"Frac < 0.05: {frac_005:.2f} (exp: 0.05)\n"
            f"Frac < 0.10: {frac_010:.2f} (exp: 0.10)"
        )
        ax_h.text(0.97, 0.95, summary_text, transform=ax_h.transAxes,
                  fontsize=7, verticalalignment="top", horizontalalignment="right",
```

```
                bbox=dict(boxstyle="round,pad=0.3", facecolor="wheat", alpha=0.8))

        # QQ plot
        ax_q = axes[i, 1]
        p_sorted = np.sort(p)
        expected = (np.arange(1, m + 1) - 0.5) / m

        ax_q.scatter(expected, p_sorted, s=20, alpha=0.7, color="#4393c3",
                     edgecolor="white", linewidth=0.3)
        ax_q.plot([0, 1], [0, 1], "r--", linewidth=1)

        # Add 95% confidence band for order statistics under Uniform
        from scipy.stats import beta as beta_dist
        lower = beta_dist.ppf(0.025, np.arange(1, m + 1), m - np.arange(1, m + 1) + 1)
        upper = beta_dist.ppf(0.975, np.arange(1, m + 1), m - np.arange(1, m + 1) + 1)
        ax_q.fill_between(expected, lower, upper, alpha=0.15, color="red",
                          label="95% pointwise band")

        ax_q.set_xlabel("Expected (Uniform)")
        ax_q.set_ylabel("Observed p-value")
        ax_q.set_title(f"{label}     QQ Plot")
        ax_q.legend(fontsize=7)
        ax_q.set_xlim(-0.02, 1.02)
        ax_q.set_ylim(-0.02, 1.02)

    plt.tight_layout()
    plt.savefig(save_path, bbox_inches="tight", dpi=150)
    plt.show()


# Need scipy.stats for KS test
from scipy import stats

# Collect all p-value sets
pval_dict = {}
# Rush F1 engines
pval_dict["Rush F1     Z-test"] = res_rush_f1_z["pval"].values
pval_dict["Rush F1     Logistic"] = res_rush_f1_logit["pval"].values
pval_dict["Rush F1     Permutation"] = res_rush_f1_perm["pval"].values
# Rush F2
pval_dict["Rush F2     Z-test"] = res_rush_f2_z["pval"].values
# Cash and Tip
pval_dict["Cash F1     Z-test"] = res_cash_f1["pval"].values
pval_dict["Tip Propensity F2     Z-test"] = res_tip_f2["pval"].values

comprehensive_pval_panel(pval_dict, "fig_6_3_pvalue_diagnostics.png")


print(f"{'Analysis':<30s} {'m':>4s} {'KS stat':>8s} {'KS p':>7s} "
      f"{'<0.05':>6s} {'<0.10':>6s} {'Interpretation':<25s}")
print("-" * 95)

for label, pvals in pval_dict.items():
    p = np.asarray(pvals, float)
```

```
p = p[np.isfinite(p)]
m = len(p)
ks_stat, ks_pval = stats.kstest(p, "uniform")
f05 = np.mean(p < 0.05)
f10 = np.mean(p < 0.10)

if ks_pval > 0.10:
    interp = "Consistent with null"
elif f05 > 0.20:
    interp = "Strong signal (left-skewed)"
elif f05 > 0.10:
    interp = "Moderate signal"
else:
    interp = "Mild departure"

print(f"{label:<30s} {m:>4d} {ks_stat:>8.3f} {ks_pval:>7.3f} " f"{f05:>6.2f} {f10:>6.2
```